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TT4D: A Pipeline and Dataset for Table Tennis 4D Reconstruction From Monocular Videos

TT4D：单目视频乒乓球 4D 重建的流程和数据集

Conference: Proceedings of the 34th ACM International Conference on Multimedia; November 10–14, 2026; Rio de Janeiro, BrazilProceedings of the 34th ACM International Conference on Multimedia (MM ' 26), November 10–14, 2026, Rio de Janeiro, BrazilDOI: 10.1145/3767308.3835475ISBN: 979-8-4007-2213-4/2026/113214CCS: Computing methodologies TrackingCCS: Computing methodologies Video segmentationCCS: Computing methodologies 3D imagingCCS: Computing methodologies Trajectory modeling

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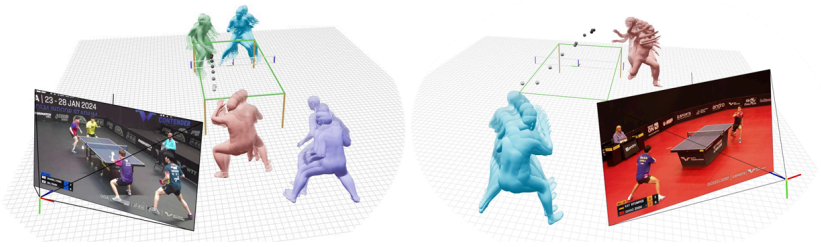


Figure 1 .

图1。

“Lift-First Pipeline" to generate TT4D, a 140+ hour dataset that recovers camera parameters, 3D ball positions, ball spin, and 3D human meshes over time from general-view broadcast videos.

“先提升后分段”（Lift-First）流程生成了 TT4D 数据集，时长超过 140 小时；该流程从一般视角的比赛转播视频中恢复随时间变化的相机参数、球的三维位置、球的旋转和三维人体网格。

We lift full 2D ball tracks to 3D trajectories, bypassing fragile 2D-based time segmentation used in traditional methods.

我们将**完整的二维球轨迹**提升为三维轨迹，从而绕开传统方法中较不稳健的二维时间分段步骤。

Two side-by-side 3D visualizations demonstrating the multimodal output of the Lift-First Pipeline.

两个并排的 3D 可视化展示了 Lift-First Pipeline 的多模态输出。

Each visualization shows a 2D broadcast video frame projected into a 3D grid space, originating from a camera coordinate axis.

每个可视化效果都显示了投影到 3D 网格空间中的 2D 转播视频帧，该帧源自摄像机坐标轴。

Within the 3D space, a green wireframe table tennis table is visible.

在 3D 空间内，可以看到一张绿色线框乒乓球桌。

A sequence of grey spheres maps the 3D ball positions across the table.

一系列灰色球体映射了球台上的 3D 球位置。

Colored 3D human meshes are rendered with multiple overlapping translucent poses to illustrate their movement over time.

彩色 3D 人体网格通过多个重叠的半透明姿态进行渲染，以说明他们随时间的移动。

The left visualization depicts a doubles match with four human meshes, while the right visualization depicts a singles match with two human meshes.

左侧可视化描绘了与四个人类网格的双打比赛，而右侧可视化描绘了与两个人类网格的单打比赛。

Abstract.

摘要

We present TT4D , a large-scale, high-fidelity table tennis dataset.

本文提出 TT4D，一个大规模、高保真的乒乓球数据集。

It provides 140+ hours of reconstructed singles and doubles gameplay from monocular broadcast videos, featuring annotations like high-quality camera calibrations, precise 3D ball positions, ball spin, time segmentation, and 3D human meshes over time.

它从单目转播视频中重建了超过 140 小时的单打和双打比赛，并提供高质量相机标定、精确三维球位置、球旋转、时间分段以及随时间变化的三维人体网格等标注。

This rich data provides a new foundation for virtual replay, in-depth player analysis, and robot learning.

这些丰富的数据为虚拟回放、深入的球员分析和机器人学习提供了新的基础。

The dataset’s combination of scale and precision is achieved through a novel reconstruction pipeline.

数据集兼具规模与精度，得益于一套新的重建流程。

Prior methods first partition a game sequence into individual shot segments using the 2D ball track, and only then attempt reconstruction.

已有方法先根据二维球轨迹将比赛序列分割为各个击球片段，再尝试进行三维重建。

However, 2D-based time segmentation collapses under occlusion and varied camera viewpoints, preventing reliable reconstruction.

然而，基于二维轨迹的时间分段在遮挡和不同机位下容易失效，从而妨碍可靠重建。

We invert this paradigm by first lifting the entire unsegmented 2D ball track to 3D through a learned lifting network.

我们将这一顺序反转：先通过学习到的提升网络，将整段未经分割的二维球轨迹提升到三维。

This 3D trajectory then allows us to reliably perform time segmentation.

随后，利用得到的三维轨迹可靠地完成时间分段。

The learned lifting network also infers the ball’s spin, handles unreliable ball detections, and successfully reconstructs the ball trajectory in cases of high occlusion.

该提升网络还推断球的旋转、处理不可靠的检测结果，并在严重遮挡时成功重建球轨迹。

This lift-first design is necessary, as our pipeline is the only method capable of reconstructing table tennis gameplay from general-view broadcast monocular videos.

这种“先提升”的设计至关重要：作者认为，本文流程是唯一能从一般视角的单目转播视频中重建乒乓球比赛的方法。

We demonstrate the dataset’s fidelity through two downstream tasks: estimating the racket’s pose & velocity at impact, and training a generative model of competitive rallies.

我们通过两个下游任务展示数据集的保真度：估计击球瞬间的球拍姿态与速度，以及训练竞技回合的生成模型。

Keywords:
关键词：

Table Tennis, 4D Reconstruction, Trajectory Estimation, Ball Spin Estimation, Sports Analytics, Time Segmentation

乒乓球、4D 重建、轨迹估计、球旋转估计、运动分析、时间分段

[†]

1. Introduction

1. 引言

Online platforms host a vast and growing collection of high-quality competitive sports footage.

在线平台拥有大量且不断增长的高质量竞技体育镜头。

This abundance of broadcast video makes 4D monocular-view reconstruction a scalable task enabling virtual replay, player analytics, and robot learning.

如此丰富的转播视频使得 4D 单目视图重建成为一项可扩展的任务，支持虚拟重播、球员分析和机器人学习。

Table tennis, in particular, serves as a challenging testbed due to its high-speed and dynamic nature.

尤其是乒乓球运动，由于其高速和动态的特性，成为一个具有挑战性的测试平台。

A complete analysis goes beyond human mesh recovery and ball position reconstruction: it includes estimating ball spin, which strongly influences both flight (via the Magnus effect) and bounce behavior.

完整的分析超出了人体网格恢复和球位置重建的范围：它包括估计球的旋转，这强烈影响飞行（通过马格努斯效应）和弹跳行为。

Extracting these signals at scale from broadcast video is difficult: the ball is small, moves rapidly, and is routinely occluded by players.

从转播视频中大规模提取这些信号很困难：球很小，移动迅速，并且经常被球员遮挡。

This constant occlusion makes time segmentation, the task of identifying the exact moment of a hit, the biggest challenge for reconstruction.

这种持续的遮挡使得时间分割（识别准确的击中时刻的任务）成为重建的最大挑战。

Existing pipelines (Etaat et al., 2025 ; Gossard et al., 2025 ; Ertner et al., 2024 ; Kienzle et al., 2025 ; Kienzle et al., 2026) follow a 2D-based strategy: first partition the sequence into shot segments using the 2D ball track, then lift each segment to 3D.

现有的流程（Etaat 等人，2025；Gossard 等人，2025；Ertner 等人，2024；Kienzle 等人，2025；Kienzle 等人，2026）遵循基于 2D 的策略：首先使用 2D 球轨迹将序列划分为击球段，然后将每个段提升到 3D。

Automated 2D-based segmentation, as in LATTE-MV (Etaat et al., 2025) and TT3D (Gossard et al., 2025) , often fails when the 2D track is interrupted by occlusions or corrupted by misdetections.

基于自动 2D 的分割，如 LATTE-MV（Etaat 等人，2025）和 TT3D（Gossard 等人，2025），当 2D 轨道因遮挡而中断或因误检测而损坏时，通常会失败。

Manual segmentation (Kienzle et al., 2025 ; Kienzle et al., 2026) allows for precise benchmarking but is not scalable, feasible only for small test sets.

手动分割（Kienzle et al., 2025；Kienzle et al., 2026）允许精确的基准测试，但不可扩展，仅适用于小型测试集。

In this work, we introduce the Lift-First Pipeline , which fundamentally reverses this logic.

在这项工作中，我们引入了 Lift-First Pipeline ， 它从根本上扭转了这一逻辑。

Our pipeline directly lifts the entire unsegmented 2D ball track to a continuous 3D trajectory using a learned model.

我们的流程使用学习模型直接将整个未分段的 2D 球轨迹提升为连续的 3D 轨迹。

Only then do we perform time segmentation in the unambiguous 3D domain.

只有这样我们才能在明确的 3D 域中执行时间分段。

Once this 3D trajectory is available, hit and bounce events can be reliably identified.

一旦获得该 3D 轨迹，就可以可靠地识别撞击和弹跳事件。

Our Lift-First Pipeline is enabled by our core technical contribution: a novel Full-Sequence Lifting Network .

Lift-First 流程依赖本文的核心技术贡献：全序列三维提升网络（Full-Sequence Lifting Network）。

By taking a 3D-first approach, we can process long and complex unsegmented gameplay for the first time.

通过采用 3D 优先的方法，我们可以首次处理冗长且复杂的未分段比赛过程。

This network is trained on a massive-scale synthetic dataset of 3 million rallies.

该网络在 300 万个回合的大规模合成数据集上进行训练。

We summarize our contributions as follows:

我们的贡献总结如下：

- The “Lift-First" Reconstruction Pipeline: A new paradigm that decouples 3D reconstruction from fragile 2D-based time segmentation by first lifting the entire unsegmented sequence to 3D.

  “Lift-First”重建流程：一种新范例，通过首先将整个未分段序列三维提升到 3D，将 3D 重建与脆弱的基于 2D 的时间分段解耦。

- A Novel Full-Sequence Lifting Network: The core technical method, enabled by a 3M points synthetic dataset.

  全序列三维提升网络：本文的核心方法，由包含 300 万个得分回合的合成数据集支撑。

  The network is the first to process unsegmented rallies and outputs the full 3D trajectory and dense per-frame 3D spin vectors.

  该网络首次处理未经击球分段的完整回合，同时输出完整三维轨迹及逐帧三维旋转向量。

- The TT4D Dataset: A 140+ hour dataset generated with our pipeline, featuring precise 3D ball trajectories, 3D human meshes, and two annotations previously unavailable at scale: dense ball spin and robust 3D-derived time segmentations.

TT4D 数据集：使用我们的流程生成的 140 多个小时的数据集，具有精确的 3D 球轨迹、3D 人体网格以及以前无法大规模使用的两个标注：密集的球旋转和强大的 3D 衍生时间分段。

- Novel Downstream Applications: We demonstrate our dataset’ s high fidelity by (a) introducing a new racket stroke estimation method that recovers the racket’ s pose and velocity at impact and (b) training a generative Flow Matching model on competitive gameplay.

  新颖的下游应用：我们通过以下方式展示了数据集的高保真度：(a) 引入一种新的球拍击球估计方法，该方法可以恢复球拍撞击时的姿态和速度；(b) 训练竞技比赛中的生成流匹配模型。

We will release the TT4D dataset, paving the way for new applications in computational sports science and robotics (Etaat et al., 2025 ; Wang et al., 2012 ; Wang et al., 2017) .

我们将发布 TT4D 数据集，为计算运动科学和机器人学的新应用铺平道路（Etaat 等，2025；Wang 等，2012；Wang 等，2017）。

2. Related Work

2.相关工作

2D ball tracking in table tennis is challenging due to its small size, fast motion, frequent occlusions, and motion blur.

乒乓球运动中的 2D 球跟踪由于其尺寸小、运动快、频繁遮挡和运动模糊而具有挑战性。

Recent methods rely on deep detectors (Huang et al., 2019 ; Zandycke and Vleeschouwer, 2019 ; Komorowski et al., 2019 ; Sun et al., 2020) , with the Multiple-Input Multiple-Output (MIMO) formulation from TrackNetV2 (Sun et al., 2020) being a key breakthrough.

最近的方法依赖于深度检测器（Huang et al., 2019；Zandycke and Vleeschouwer, 2019；Komorowski et al., 2019；Sun et al., 2020），TrackNetV2 的多输入多输出（MIMO）公式（Sun et al., 2020）是一个关键突破。

This MIMO strategy has been adopted by subsequent works using different backbones (Tarashima et al., 2023 ; Liu and Wang, 2022 ; Chen and Wang, 2024 ; Kienzle et al., 2024 ; Kienzle et al., 2026) .

后续工作在采用不同主干网络时，也沿用了这一 MIMO 策略（Tarashima 等人，2023；Liu 和 Wang，2022；Chen 和 Wang，2024；Kienzle 等人，2024；Kienzle 等人，2026）。

Attention mechanisms have also been incorporated to enhance temporal feature fusion (Hu et al., 2018 ; Gossard et al., 2026 ; Raj et al., 2025) .

还引入了注意力机制来增强时间特征融合（Hu et al., 2018；Gossard et al., 2026；Raj et al., 2025）。

Methods for 3D trajectory lifting can be split into physics-based optimization and learned networks.

将二维轨迹提升到三维的方法可分为基于物理的优化方法和学习型网络。

Optimization-based methods like TT3D (Gossard et al., 2025) , LATTE-MV (Etaat et al., 2025) and MonoTrack (Liu and Wang, 2022) minimize reprojection error.

基于优化的方法，如 TT3D（Gossard 等人，2025）、LATTE-MV（Etaat 等人，2025）和 MonoTrack（Liu 和 Wang，2022）可最大限度地减少重投影误差。

These non-learning methods are inherently limited to the “Traditional Pipeline" and hence do not scale.

这些非学习方法本质上仅限于“传统流程”，因此无法扩展。

Their optimization is already unstable for single, shot segments; extending them to a “3D-first" approach that jointly solves for the trajectory, all unknown bounce points, and all unknown hit-points is computationally infeasible.

它们的优化对于单个击球段来说已经不稳定；将它们扩展到“3D优先”的方法来联合求解轨迹、所有未知的弹跳点和所有未知的击球点在计算上是不可行的。

This necessitates a learning-based approach to make the Lift-First Pipeline viable.

这就需要一种基于学习的方法来使提升优先流程变得可行。

Recent learned approaches train a network to lift 2D tracks to 3D trajectories.

最近学习的方法训练网络将 2D 轨道提升到 3D 轨迹。

SynthNet (Ertner et al., 2024) and (Ponglertnapakorn and Suwajanakorn, 2025) both tackle this for tennis.

SynthNet（Ertner 等人，2024）和（Ponglertnapakorn 和 Suwajanakorn，2025）都在网球领域解决了这个问题。

Most relevant is the work of Kienzle et al. (2025) , which proposes a transformer that lifts single shot segments from 2D to 3D with zero-shot generalization from synthetic to real data.

最相关的是 Kienzle 等人（2025）的工作：它使用 Transformer 将单个击球片段从二维提升到三维，并实现从合成数据到真实数据的零样本泛化。

This was extended in (Kienzle et al., 2026) and serves as the basis for our network.

这在（Kienzle et al., 2026）中得到了扩展，并作为我们网络的基础。

However, all these methods still depend on the “Traditional Pipeline" and its dependence on fragile 2D-based time segmentation.

然而，所有这些方法仍然依赖于“传统流程”及其对脆弱的基于 2D 的时间分割的依赖。

We bypass this limitation and develop a network that can lift full rallies from 2D to 3D.

我们绕过了这个限制，开发了一个可以将完整回合从 2D 提升到 3D 的网络。

Spin estimation methods include indirect inference from trajectories or direct logo tracking (Tebbe et al., 2020) .

旋转估计方法包括从轨迹间接推断或直接标识跟踪（Tebbe 等人，2020）。

Direct tracking has been improved with custom dot patterns (Gossard et al., 2023) and event cameras that mitigate motion blur (Gossard et al., 2024 ; Nakabayashi et al., 2024) .

通过自定义点图案（Gossard 等人，2023）和减轻运动模糊的事件相机（Gossard 等人，2024；Nakabayashi 等人，2024），直接跟踪得到了改进。

These hardware-specific methods are complemented by works that classify spin from player stroke motion (Kulkarni and Shenoy, 2021 ; Fujihara et al., 2025) .

这些特定于硬件的方法得到了对球员击球运动中的旋转进行分类的工作的补充（Kulkarni 和 Shenoy，2021；Fujihara 等人，2025）。

Most recently, Kienzle et al. (2025) ; Kienzle et al. (2026) showed that 2D-3D lifting transformers can also regress the initial 3D spin vector.

Kienzle 等人（2025、2026）进一步表明，二维到三维提升 Transformer 也可以回归球的初始三维旋转向量。

We adapt this to predict per-frame spin for an entire unsegmented rally.

我们调整它来预测整个未分段回合的每帧旋转。

Specialized table-tennis datasets have only emerged recently.

专门的乒乓球数据集最近才出现。

Blurball (Gossard et al., 2026) provided blur-aware 2D annotations, while TT3D (Gossard et al., 2025) offered precise 3D trajectories from a multi-camera setup.

Blurball（Gossard 等人，2026）提供了模糊感知的 2D 标注，而 TT3D（Gossard 等人，2025）则通过多摄像机设置提供精确的 3D 轨迹。

Synthetic datasets of individual shot segments have been used for model training (Kienzle et al., 2025 ; Kienzle et al., 2026) , along with small-scale real-world sets with topspin/backspin annotations (Kienzle et al., 2025 ; Kienzle et al., 2026) .

各个击球片段的合成数据集已用于模型训练（Kienzle 等人，2025；Kienzle 等人，2026），以及带有上旋/下旋标注的小规模现实世界数据集（Kienzle 等人，2025；Kienzle 等人，2026）。

TTNet (Voeikov et al., 2020) and P2ANet (Bian et al., 2024) primarily focus on event spotting and fine-grained action detection within broadcast or multi-task video contexts.

TTNet（Voeikov 等人，2020）和 P2ANet（Bian 等人，2024）主要关注广播或多任务视频环境中的事件发现和细粒度动作检测。

Most similar to our TT4D dataset in scale is LATTE-MV (Etaat et al., 2025) , which reconstructed 26 hours of gameplay using the “Traditional Pipeline”.

在规模上与我们的 TT4D 数据集最相似的是 LATTE-MV（Etaat 等人，2025），它使用“传统流程”重建了 26 小时的比赛时长。

Our TT4D dataset, enabled by our Lift-First Pipeline, surpasses this by an order of magnitude.

利用 Lift-First 流程生成的 TT4D 数据集，在规模上高出一个数量级。

Beyond scale, TT4D provides higher-fidelity data, using an improved camera calibration method and providing realistic 3D trajectories, in contrast to LATTE-MV’s simplified parabolic fits.

除规模外，TT4D 还通过改进相机标定提供更高保真度的数据，并重建更真实的三维轨迹，而 LATTE-MV 使用简化的抛物线拟合。

Crucially, it is the first to provide two key annotations at this scale: dense, per-frame 3D spin and robust 3D-derived time segmentation, which are reliable even when 2D occlusions break prior methods.

至关重要的是，它是第一个在这种规模上提供两个关键标注的技术：密集的每帧 3D 旋转和强大的 3D 衍生时间分段，即使 2D 遮挡破坏了先前的方法，这些标注也是可靠的。

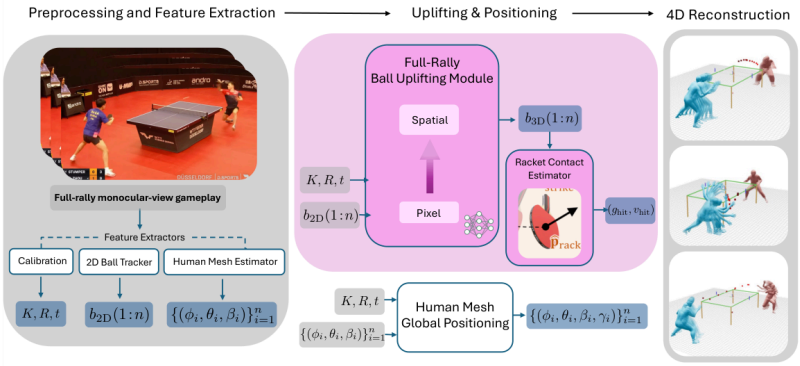


Figure 2 .

图2.

A visual outline of our proposed Lift-First Pipeline.

我们提议的“提升优先”流程的视觉轮廓。

Instead of depending on challenging and noisy temporal segmentations of the video sequence before attempting the 3D uplifting of the trajectory, we invert this logic by lifting the entire sequence to 3D first, which makes subsequent temporal segmentation and refinement in the 3D domain a simple and robust task.

在尝试 3D 提升轨迹之前，我们不是依赖于视频序列的具有挑战性和含噪的时间分割，而是首先通过将整个序列三维提升到 3D 来反转此逻辑，这使得 3D 域中的后续时间分割和细化成为一项简单而稳健的任务。

A block diagram illustrating the three stages of the Lift-First Pipeline: Preprocessing and Feature Extraction, Uplifting and Positioning, and 4D Reconstruction.

框图说明了 Lift-First 流程的三个阶段：预处理和特征提取、提升和定位以及 4D 重建。

In the first stage, a broadcast video of a table tennis match is processed by three feature extractors: Calibration (outputting K, R, t), 2D Ball Tracker (outputting $b_{2D}(1:n)$), and Human Mesh Estimator (outputting $\{(\phi_i, \theta_i, \beta_i)\}_{i=1}^n$).

在第一阶段，乒乓球比赛的转播视频由三个特征提取器处理：校准（输出K, R, t）、2D Ball Tracker（输出$b_{2D}(1:n)$）和人体网格估计器（输出$\{(\phi_i, \theta_i, \beta_i)\}_{i=1}^n$）。

In the second stage, a Full-Rally Ball Uplifting Module takes the 2D ball track and calibration data through a neural network to produce a 3D trajectory $b_{3D}(1:n)$, which then feeds into a Racket Contact Estimator to estimate racket hit parameters.

第二阶段，全回合球轨迹提升模块将二维球轨迹和标定信息送入神经网络，生成三维轨迹 $b_{3D}(1:n)$；球拍接触估计器据此估计击球参数。

Concurrently, a Human Mesh Global Positioning module combines the mesh parameters and calibration data to output globally positioned meshes $\{(\phi_i, \theta_i, \beta_i, \gamma_i)\}_{i=1}^n$.

同时，人体网格全局定位模块结合网格参数和校准数据来输出全局定位网格$\{(\phi_i, \theta_i, \beta_i, \gamma_i)\}_{i=1}^n$。

The final stage displays three 3D rendered frames of the resulting 4D reconstruction, showing two human meshes interacting with the ball trajectory at a table tennis table.

最后阶段显示最终 4D 重建的三个 3D 渲染帧，显示两个人体网格与乒乓球台上的球轨迹相互作用。

3. Methodology: The Lift-First Pipeline

3. 方法论：提升优先流程

3.1. Terminology

3.1.术语

We first define terminology used throughout the paper:

我们首先定义整篇论文中使用的术语：

- A Segment starts when one player hits the ball and ends when another player hits the ball.

  当一名球员击球时，一个分段开始，当另一名球员击球时结束。

- A Point starts with a serve and ends when the ball is first out of play.

  一个得分回合（Point）从发球开始，到球首次退出比赛状态时结束。

  Segments partition the point into disjoint time intervals.

  各个击球片段将一个回合划分为互不重叠的时间区间。

- A Game is a sequence of points.

  一局（Game）由一系列得分回合组成。

  It is complete when one team achieves the winning score.

  当一方达到获胜分数时，该局结束。

- (2D/3D)-based Time Segmentation partitions a point into segments using 2D or 3D information.

  基于二维或三维的时间分段，是利用相应空间信息将一个回合划分为各个击球片段。

-

4D Reconstruction recovers camera parameters, 3D ball positions, ball spin, and 3D human meshes over time .

4D 重建可随时间恢复摄像机参数、3D 球位置、球旋转和 3D 人体网格。

3.2. Pipeline Overview

3.2.流程概览

Conventional table tennis reconstruction pipelines first perform 2D-based time segmentation of a point and then reconstruct each segment independently (Etaat et al., 2025 ; Gossard et al., 2025) .

传统乒乓球重建流程先利用二维信息对一个回合进行时间分段，再逐段独立重建（Etaat 等，2025；Gossard 等，2025）。

However, this time segmentation scheme is highly sensitive to occlusions and missing detections, limiting scalability.

然而，这种时间分割方案对遮挡和缺失检测高度敏感，限制了可扩展性。

We instead adopt a Lift-First Pipeline (Fig. 2).

相反，我们采用了 Lift-First Pipeline（提升优先流程）（图 2）。

Rather than segmenting first, we lift the entire unsegmented point to 3D and perform time segmentation and annotation directly in the 3D domain, where the ball trajectory signal is unambiguous.

本文先将整个未分段回合提升到三维，再利用三维空间中更明确的球路信号进行时间分段和标注。

The pipeline consists of four stages:

该流程由四个阶段组成：

(1)

Data Acquisition and Preprocessing (Section 3.3): clipping the points of a game from broadcast footage, calibrating cameras, extracting 2D ball tracks, and estimating 3D human meshes (Loper et al., 2015) .

数据采集与预处理（第 3.3 节）：从转播视频中裁出各个得分回合，标定相机，提取二维球轨迹并估计三维人体网格（Loper 等，2015）。

(2)

Full 3D Lifting (Section 3.4): a transformer-based network predicts dense 3D ball trajectories and per-frame spin for the full point from the 2D track.

全 3D 三维提升（第 3.4 节）：基于Transformer的网络预测 2D 轨迹完整回合的密集 3D 球轨迹和每帧旋转。

(3)

3D-Domain Annotation (Section 3.5): segments, hit points, bounces, and racket-contact parameters are computed from the reconstructed 3D trajectory.

3D 域标注（第 3.5 节）：根据重建的 3D 轨迹计算段、击球点、弹跳和球拍接触参数。

(4)

Filtering and Curation (Section 3.6): 2D and 3D consistency checks ensure that all retained trajectories are visually and physically plausible.

过滤和管理（第 3.6 节）：2D 和 3D 一致性检查确保所有保留的轨迹在视觉和物理上都是合理的。

3.3. Stage 1: Data Acquisition and Preprocessing

3.3.第一阶段：数据采集和预处理

Our pipeline begins with raw, multi-hour uncut table tennis videos of full games.

我们的流程从完整比赛的原始、多小时未剪辑的乒乓球视频开始。

We apply a two-stage clipping process.

我们应用两阶段裁剪过程。

The first stage splits the games into individual points by detecting scoreboard changes using YOLO (Varghese and M., 2024) and PaddleOCR (Cui et al., 2025) .

第一阶段使用 YOLO（Varghese 和 M.，2024）和 PaddleOCR（Cui 等，2025）检测记分牌变化，将比赛切分为各个得分回合。

The second stage identifies the approximate start and end of the point using 2D ball track oscillations using a ball tracker.

第二阶段根据跟踪器获得的二维球轨迹的往返变化，确定回合的大致起止时刻。

We then process all clips to detect and remove duplicated frames, a common artifact in broadcast video that corrupts trajectory estimation.

然后，我们处理所有剪辑以检测并删除重复的帧，这是转播视频中常见的伪影，会破坏轨迹估计。

Our method uses SSIM (SSIM) (Wang et al., 2004) to identify these frames.

我们的方法使用 SSIM (SSIM) (Wang et al., 2004) 来识别这些帧。

For each resulting valid clip, we extract the following information:

对于每个生成的有效剪辑，我们提取以下信息：

- Camera calibration: We follow TT3D (Gossard et al., 2025) , solving a Perspective- n -Point problem from table corners with unknown focal length, and improving robustness through enhanced table segmentation and temporal filtering.

  相机标定：沿用 TT3D（Gossard 等，2025）的方法，根据球台角点求解焦距未知的透视 n 点问题，并通过改进球台分割及时间滤波提高鲁棒性。

- 2D ball detections: TrackNetV3 (Chen and Wang, 2024) is applied without its inpainting module.

  二维球检测：使用 TrackNetV3（Chen 和 Wang，2024），但移除其轨迹补全模块。

- 3D human meshes: We use 4DHumans (Goel et al., 2023) and align the meshes to the world frame.

  3D 人体网格：我们使用 4DHumans（Goel 等人，2023）并将网格与世界坐标系对齐。

Additional implementation and parameter details are provided in the supplementary material.

补充材料中提供了其他实现和参数详细信息。

3.4. Stage 2: Full-Sequence Lifting Network

3.4.第二阶段：全序列三维提升网络

The central component of our pipeline is a transformer-based Full-Sequence Lifting Network.

本流程的核心是基于 Transformer 的全序列三维提升网络。

For each point consisting of N frames, it processes the sequence of 2D ball detections $\{\vec{r}_{2D}(t_n)\}_{n=0}^{N-1}$, their corresponding timestamps $\{t_n\}_{n=0}^{N-1}$, and a set of 2D table keypoints $\{\vec{k}_i\}_{i=1}^{13}$

that are derived from the camera calibration.

对于包含 N 帧的每个回合，网络处理二维球检测序列 $\{\vec{r}_{2D}(t_n)\}_{n=0}^{N-1}$、对应时间戳 $\{t_n\}_{n=0}^{N-1}$，以及由相机标定获得的二维球台关键点 $\{\vec{k}_i\}_{i=1}^{13}$。

The network infers the 3D trajectory $\{\vec{r}_{3D}(t_n)\}_{n=0}^{N-1}$ and 3D spin vectors $\{\vec{\omega}(t_n)\}_{n=0}^{N-1}$ for each frame.

网络为每一帧推断三维轨迹 $\{\vec{r}_{3D}(t_n)\}_{n=0}^{N-1}$ 和三维旋转向量 $\{\vec{\omega}(t_n)\}_{n=0}^{N-1}$。

Our network is built upon the baseline lifting model from Kienzle et al. (2026) , which demonstrated strong generalization to real videos despite being trained solely on synthetic data.

我们的网络以 Kienzle 等人（2026）的三维提升模型为基础；该模型虽然仅用合成数据训练，却能很好地泛化到真实视频。

We retain its key innovations, such as using Rotary Positional Embeddings (RoPE) (Su et al., 2024) based on exact timestamps to handle varying frame rates and missing 2D ball detections (Kienzle et al., 2026) .

我们保留了其关键设计，例如基于精确时间戳的旋转位置编码 RoPE（Su 等，2024），以应对不同帧率和二维球检测缺失（Kienzle 等，2026）。

However, the baseline model is designed for a “Traditional Pipeline”: it processes isolated, pre-segmented shots, predicts only a single initial spin vector per segment, and handles missing 2D ball detections by simply discarding them.

但该基线面向“传统流程”：只处理独立且预先分割好的击球片段，每段仅预测一个初始旋转向量，并直接丢弃缺少二维球检测的帧。

This is insufficient for our “Lift-First Pipeline,” which must process unsegmented points of arbitrary length, generate dense spin estimates, and actively reconstruct occluded detections to enable precise 3D-based temporal segmentation.

这些能力不足以支撑“先提升流程”：后者必须处理任意长度、未经分割的回合，生成逐帧旋转估计，并主动恢复遮挡处的状态，以支持精确的三维时间分段。

Therefore, we introduce three key contributions to solve this: a massive-scale synthetic training dataset of full points, architectural extensions for modeling dense spin, and an interpolation token that enables predictions for missed detections.

为此，我们引入三项关键改进：包含完整回合的大规模合成训练数据集、用于预测逐帧旋转的网络结构扩展，以及支持漏检位置预测的插值词元。

Synthetic Dataset.

合成数据集。

To learn the dynamics of continuous play, we require training data that reflects the complexity of full, unsegmented points, not just isolated segments as is done in (Kienzle et al., 2025 ; Kienzle et al., 2026) .

为学习连续比赛的运动规律，训练数据必须体现完整、未经分割回合的复杂性，而不能只包含 Kienzle 等人（2025、2026）使用的独立片段。

We therefore generate a new massive-scale synthetic dataset of 3 million points using the MuJoCo (Todorov et al., 2012) physics simulation environment.

因此，我们利用 MuJoCo 物理仿真环境（Todorov 等，2012）生成了包含 300 万个回合的新合成数据集。

We develop an iterative “stitching” algorithm.

我们开发了一种迭代拼接算法。

We first simulate a pre-serve ball toss; at its apex, we query a large data pool of initial conditions of serves from (D’Ambrosio et al., 2025 ; Kienzle et al., 2026) for the closest match in position.

首先模拟发球前的抛球；在抛球最高点，从已有发球初始条件的大型数据池中查询位置最接近的样本（D’Ambrosio 等，2025；Kienzle 等，2026）。

This serve is rolled out and from its terminal state we again query a large data pool of standard segments.

随后模拟该发球过程，并根据其终止状态再次查询标准击球片段的大型数据池。

By iteratively matching and stitching these sampled trajectory segments, we produce continuous, physically plausible sequences that enable training our network on full unsegmented points.

通过反复匹配并拼接采样得到的轨迹片段，我们生成连续且符合物理规律的序列，从而能够用完整、未经分割的回合训练网络。

Dense Spin Predictions.

逐帧旋转预测。

We adapt the baseline network architecture to exploit this new continuous data.

我们调整基线网络结构，以利用这些新的连续数据。

The architecture is illustrated in the supplementary material.

网络结构见补充材料。

The baseline model uses a learnable "spin token" to aggregate information and predicts a single initial spin vector $\vec{\omega}(t_0)$ for the input segment.

基线使用一个可学习的"旋转词元"聚合信息，为输入片段预测单个初始旋转向量 $\vec{\omega}(t_0)$。

This is no longer feasible for processing full points, as the segments in the point are not known.

处理完整回合时，各击球片段尚未确定，因此这一设计不再适用。

We therefore remove this specialized token entirely.

所以，我们完全移除了这一专用词元。

Instead, we modify the network to predict spin in a dense, per-frame manner by applying a small MLP head to every output token of the transformer.

取而代之的是：在 Transformer 的每个输出词元上接入一个小型 MLP 预测头，以密集、逐帧的方式预测旋转。

To encourage the network to learn robust trajectory and spin dynamics for points of arbitrary length, we sample a subsequence (with a random length between 20 and 250 frames) from each full point in our synthetic dataset.

为让网络学习任意长度回合中鲁棒的轨迹和旋转动力学，我们从合成数据集的每个完整回合中采样一个子序列，其长度在 20 至 250 帧之间随机选取。

Interpolation Token.

插值词元。

Ball detections are frequently missing due to occlusions in oblique views of gameplay.

在斜视比赛画面中，遮挡经常导致球检测缺失。

While the baseline architecture (Kienzle et al., 2026) simply ignores these detections, we treat the recovery of missing frames as a Masked Token Modeling (MTM) task.

基线结构（Kienzle 等，2026）直接忽略这类检测，而我们将缺失帧的恢复视为掩膜词元建模（MTM）任务。

To prevent the loss of spatial camera context when a ball is missing, we introduce a Disentangled Context Embedding (DCE) .

为防止球缺失时丢失相机的空间上下文，我们引入解耦上下文嵌入（DCE）。

The 2D ball position and the table keypoints are projected into a higher dimension vector via separate linear layers.

二维球位置与球台关键点分别经过独立的线性层，投影到更高维的向量空间。

For frames with missing ball detections, we replace only the ball vector with a learnable interpolation token, leaving the projected table keypoints intact to preserve the camera information.

对缺少球检测的帧，仅用可学习的插值词元替换球位置向量，同时保留投影后的球台关键点，以维持相机信息。

Finally, we concatenate the ball vector and table keypoint vector and apply a linear layer to obtain the final embedding for each frame.

最后，拼接球位置向量与球台关键点向量，再经过一个线性层，得到每帧的最终嵌入。

This is illustrated in the supplementary material.

补充材料给出了示意图。

To prevent polluting the valid information of successful ball detections with the interpolation tokens, we integrate Deferred Upsampling Token Attention (DUTA) (Einfalt et al., 2023) .

为避免插值词元干扰有效球检测的信息，我们引入延迟上采样词元注意力 DUTA（Einfalt 等，2023）。

DUTA applies masking in the initial transformer layer to prevent context dilution.

DUTA 在初始 Transformer 层施加注意力掩膜，防止上下文信息被稀释。

Each token is only allowed to attend to tokens representing valid detections, ensuring that the tokens representing invalid detections can gather the necessary context without diluting the information of the valid tokens.

每个词元只允许关注有效检测对应的词元；这样，无效检测词元能够获取必要上下文，同时不会稀释有效词元的信息。

During training, we randomly mask valid 2D detections using empirically-derived occlusion probabilities (discussed in supplementary materials) and compute the dense 3D reconstruction loss over the entire trajectory, including the masked frames.

训练时，我们按经验得到的遮挡概率随机遮蔽有效二维检测，并对整条轨迹（包括遮蔽帧）计算密集三维重建损失；遮挡概率的细节见补充材料。

This forces the network to internalize the underlying physical constraints of ball motion to accurately in-paint missing segments.

这促使网络学习球运动背后的物理约束，以准确补全缺失片段。

3.5. Stage 3: 3D-Domain Annotation

3.5.第 3 阶段：3D 域标注

With an unambiguous 3D ball trajectory now available, we can perform time segmentation and annotation directly in the 3D domain.

现在有了明确的 3D 球轨迹，我们可以直接在 3D 域中执行时间分段和标注。

Robust 3D-based Time Segmentation.

强大的基于 3D 的时间分段。

Our Lift-First Pipeline transforms time segmentation from a complex, 2D image-level tracking problem into an unambiguous, 1D signal analysis task in world coordinate space.

我们的 Lift-First Pipeline 将时间分割从复杂的 2D 图像级跟踪问题转换为世界坐标空间中明确的 1D 信号分析任务。

We identify hit events as the peaks and troughs in the ball’ s 3D x -coordinates, using simple time and distance heuristics to filter local noise.

我们将击球事件识别为球的 3D x 坐标中的波峰和波谷，使用简单的时间和距离启发法来过滤局部噪声。

Similarly, we label table bounces as the local minima in the z-coordinates.

类似地，将 z 坐标的局部极小值标为球触台反弹事件。

This provides the time segmentations that prior methods failed to reliably achieve.

这提供了先前方法无法可靠实现的时间分段。

Racket Stroke Estimation.

球拍击球估计。

Estimating a player’ s 3D body pose provides useful cues for anticipating ball motion, but current human-pose estimators do not reliably capture hand orientation or wrist articulation.

估计球员的 3D 身体姿态为预测球运动提供了有用的线索，但当前的人体姿态估计器无法可靠地捕获手部方向或手腕关节。

This limitation is critical: the racket orientation at impact strongly determines the outgoing ball trajectory and spin.

这一限制至关重要：击球时的球拍方向很大程度上决定了球的出球轨迹和旋转。

Prior attempts to estimate racket pose directly from video (Gao et al., 2019 ; Wang and Shi, 2013) remain insufficiently accurate or robust outside controlled laboratory settings.

已有直接从视频估计球拍姿态的方法（Gao 等，2019；Wang 和 Shi，2013），在受控实验室环境之外仍缺乏足够的准确性和鲁棒性。

Instead, we infer the racket state indirectly from the 3D ball trajectory.

相反，我们从 3D 球轨迹间接推断球拍状态。

When the ball’ s flight time is short and its spin remains within a moderate range, the two-point boundary-value problem defined by the hit position and the subsequent table bounce admits a unique physically plausible ball trajectory (Liu et al., 2012) .

当飞行时间较短且旋转处于适中范围时，以击球位置和随后触台位置定义的两点边值问题，存在唯一符合物理规律的球轨迹（Liu 等，2012）。

This provides the required ball velocity and spin immediately after impact.

这可以在击球后立即提供所需的球速和旋转。

Implementation and validation details are provided in the supplementary material.

补充材料中提供了实施和验证细节。

We use this procedure to augment our dataset with physically consistent racket-stroke parameters.

我们使用此过程通过物理上一致的球拍击球参数来扩充我们的数据集。

3.6. Stage 4: Filtering and Curation

3.6.第四阶段：筛选与整理

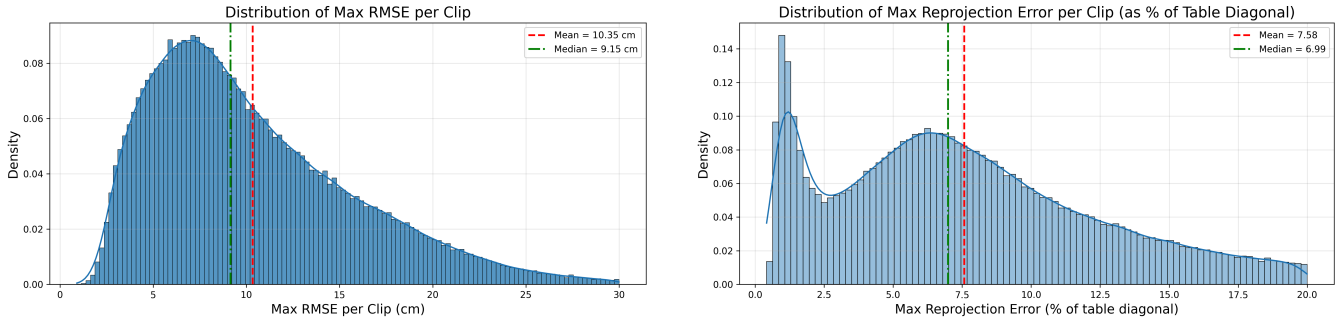


Figure 3 .

图3.

Distributions of the two quality metrics used in our Filtering pipeline.

我们的过滤流程中使用的两个质量指标的分布。

(Top) Histogram of the maximum Physics-Based ODE Fit error (RMSE) per point.

上：每个回合内最大的物理 ODE 拟合均方根误差（RMSE）分布。

(Bottom) Histogram of the maximum normalized 2D Reprojection Error per point.

下：每个回合内最大的归一化二维重投影误差分布。

The peaks in both distributions near zero demonstrate that our pipeline’ s outputs are overwhelmingly physically plausible and visually accurate.

两个分布中的峰值都接近于零，这表明我们的流程输出在物理上是合理的，并且在视觉上是准确的。

The tails of the distributions provide a clear margin for selecting reliable filtering thresholds..

分布的尾部为选择可靠的过滤阈值提供了清晰的裕度。

Two vertically stacked histograms, both showing strongly right-skewed distributions with peaks near zero and long tails extending to the right.

两个垂直堆叠的直方图，均显示强烈的右偏分布，峰值接近于零，长尾向右延伸。

The top histogram plots the maximum Physics-Based ODE Fit error (RMSE) per clip in centimeters, featuring a vertical green dash-dot line marking the median at 8.30 cm and a vertical red dashed line marking the mean at 10.77 cm.

顶部直方图绘制了每个剪辑的最大基于物理的 ODE 拟合误差 (RMSE)（以厘米为单位），其中一条垂直的绿色点划线标记了 8.30 厘米处的中值，一条垂直的红色虚线标记了 10.77 厘米处的平均值。

The bottom histogram plots the maximum normalized 2D Reprojection Error per clip as a percentage of the table diagonal, featuring a vertical green dash-dot line marking the median at 8.12 and a vertical red dashed line marking the mean at 10.28.

底部直方图绘制了每个剪辑的最大归一化 2D 重投影误差占球台对角线的百分比，其中一条垂直绿色点划线标记中位数为 8.12，一条垂直红色虚线标记平均值为 10.28。

The final stage enforces visual and physical consistency across all reconstructed rallies in order to keep only high quality reconstruction for our TT4D dataset.

最后阶段强制所有重建回合的视觉和物理一致性，以便为我们的 TT4D 数据集保留高质量的重建。

We apply one 2D-based filter and two 3D-based filters. 2D Reprojection Check.

我们应用一个基于 2D 的滤波器和两个基于 3D 的滤波器。 2D重投影检查。

We compare the reprojected 3D trajectory estimate to the original detections and normalize errors by the pixel length of the table diagonal.

我们将重投影的 3D 轨迹估计与原始检测进行比较，并通过球台对角线的像素长度对误差进行归一化。

If the maximum normalized error exceeds a strict threshold (20% of the table diagonal length), the point is rejected.

若最大归一化误差超过球台对角线长度的 20%，则剔除整个回合。

Figure 3 (b) shows the distribution of this error.

图 3 (b) 显示了该误差的分布。

Event Plausibility: We enforce game logic by analyzing the 3D path, ensuring it contains a valid sequence of events: clearly identifiable hit points and a single table bounce per segment (or two for serves).

事件合理性：我们通过分析 3D 路径来强化游戏逻辑，确保它包含有效的事件序列：清晰可辨的击球点和每段的单个球台弹跳（或发球两次）。

Physical Consistency: We assess physical consistency by fitting the ODE (ODE) constrained ball trajectory (Gossard et al., 2025) to our network’ s 3D output.

物理一致性：我们通过将 ODE (ODE) 约束球轨迹 (Gossard et al., 2025) 拟合到我们网络的 3D 输出来评估物理一致性。

This step effectively projects our prediction onto the manifold of physically possible trajectories.

这相当于将预测结果投影到物理可行轨迹构成的流形上。

If the maximum Euclidean distance between the prediction and the ODE fit exceeds a 30 cm threshold, the point is discarded as physically implausible.

若预测轨迹与 ODE 拟合轨迹之间的最大欧氏距离超过 30 cm，则认为该回合不符合物理规律并将其剔除。

The error distribution is shown in Figure 3 (a).

误差分布如图3(a)所示。

4. The TT4D Dataset

4. TT4D数据集

4.1. General Information

4.1.一般信息

The TT4D dataset is sourced from 45,946 broadcast table tennis games (best of five, first to three) from 2021-2024.

TT4D 的数据来源为 2021—2024 年的 45,946 场转播乒乓球比赛，赛制为五局三胜。

We support general stationary camera poses, singles and doubles gameplay, and video speeds of at least 25 FPS, ultimately resulting in 211,534 reconstructed points and 146 hours of gameplay.

流程支持多种固定机位、单打和双打，以及至少 25 FPS 的视频，最终得到 211,534 个重建回合，共 146 小时比赛内容。

For comparison, the LATTE-MV dataset is based on a set of 1017 games, requires a particular camera pose, only handles singles gameplay, and ultimately results in 23,782 reconstructed points and 26 hours of gameplay.

作为比较，LATTE-MV 基于 1,017 场比赛，需要特定机位且仅支持单打，最终得到 23,782 个重建回合，共 26 小时。

Our scalability stems directly from the Lift-First Pipeline.

我们的可扩展性直接源于 Lift-First Pipeline。

By lifting the full unsegmented point directly to 3D, hit points and bounces can be identified robustly in world coordinates, even under heavy occlusion.

直接将完整回合提升到三维后，即使遮挡严重，也能在世界坐标系中可靠地识别击球与触台反弹事件。

Moreover, because the 3D trajectory is reconstructed independently of human-pose tracking, the method naturally generalizes to doubles matches.

此外，由于 3D 轨迹的重建独立于人体姿态跟踪，因此该方法自然可以推广到双打比赛。

A further advantage from a game-theoretic standpoint is that the final winning shot of each point — unrecoverable in 2D-based time segmentation pipelines — is reliably reconstructed.

从比赛策略分析角度看，另一优点是能够可靠地重建每个回合的最后一拍制胜球；基于二维分段的流程难以恢复这一拍。

4.2. Filtering Effectiveness

4.2.筛选效果

We display the amount of filtered data in Table 1 .

我们在表 1 中显示了过滤后的数据量。

The first stage in the pipeline involves a two-part clipping process, which is outlined in Section 3.3 .

流程的第一阶段涉及两部分的裁剪过程，如第 3.3 节所述。

Our simple, conservative heuristic succeeds on 56.8% of the available points.

这一简单、保守的启发式流程能成功处理 56.8% 的可用回合。

Performance can be improved through refined scoreboard detection and gameplay-identification heuristics.

可以通过精细的记分板检测和游戏识别启发法来提高性能。

A small fraction of points ($\sim 5\%$) are lost due to either (1) a detected change in the camera pose or (2) calibration algorithm failure.

少量回合（$\sim 5\%$）因检测到相机位姿变化或标定失败而被剔除。

Our 3D-domain filters play a significant role in quality control.

我们的 3D 域滤波器在质量控制中发挥着重要作用。

Discrepancies in the 3D ball trajectory prediction and 2D trajectory evidence (4.3%) , measured as reprojection error, and physically implausible ball trajectories (1.4%) and make for effective filters.

三维预测与二维观测之间的重投影误差（(4.3%)），以及不符合物理规律的轨迹（(1.4%)），都提供了有效的筛选依据。

A substantial number of points are rejected due to human mesh recovery (e.g. not enough potential athletes, missing detections or improperly positioned athletes).

大量回合因人体网格恢复问题被剔除，例如检测到的潜在球员数量不足、漏检或球员定位不合理。

We also require a minimum of two segments to be reconstructed for each point, and we require an average ball visibility of at least 50%.

此外，每个回合须至少重建两个击球片段，且球的平均可见帧比例不得低于 50%。

Additional data curation details are presented in the supplementary materials.

补充材料中提供了其他数据管理详细信息。

Table 1 .

表 1.

Pipeline filtering statistics per stage.

每个阶段的流程过滤统计信息。

Broadcast videos are clipped into points at scoreboard changes.

先根据记分牌变化，将转播视频切分为得分回合片段。

We further trim clip downs to the actual gameplay, eliminating rests before and after the point.

再裁出实际比赛过程，去掉回合前后的休息画面。

Lastly, we calibrate the camera and apply consistency filters to the reconstructed ball trajectory.

最后，我们校准相机并对重建的球轨迹应用一致性滤波器。

Stage 阶段	Count 计数	% vs prev % 与上一个相比	% from start 从开始算起的百分比
Scoreboard clips 记分牌剪辑			
success 成功	714,664		100.0%
Gameplay clips 有效比赛片段			
success 成功	405,769	56.8%	56.8%
failed 失败了	308,895	43.2%	43.2%
Calibrations 校准			
success 成功	371,733	91.6%	52.0%
moving camera 移动相机	33,137	8.2%	4.6%
failed 失败了	899	0.2%	0.1%
Reconstructed 重建			
success 成功	211,534	56.9%	29.6%

invalid human pos 人体位置无效	67,442	18.1%	9.4%
not enough segments 没有足够的段	47,823	12.9%	6.7%
high reproj errors 重投影误差过大	30,937	8.3%	4.3%
high ODE fit ODE 拟合误差过大	9,736	2.6%	1.4%
low ball vis 球可见比例过低	4,261	1.1%	0.6%

4.3. Statistics and Basic Analysis

4.3.统计与基础分析

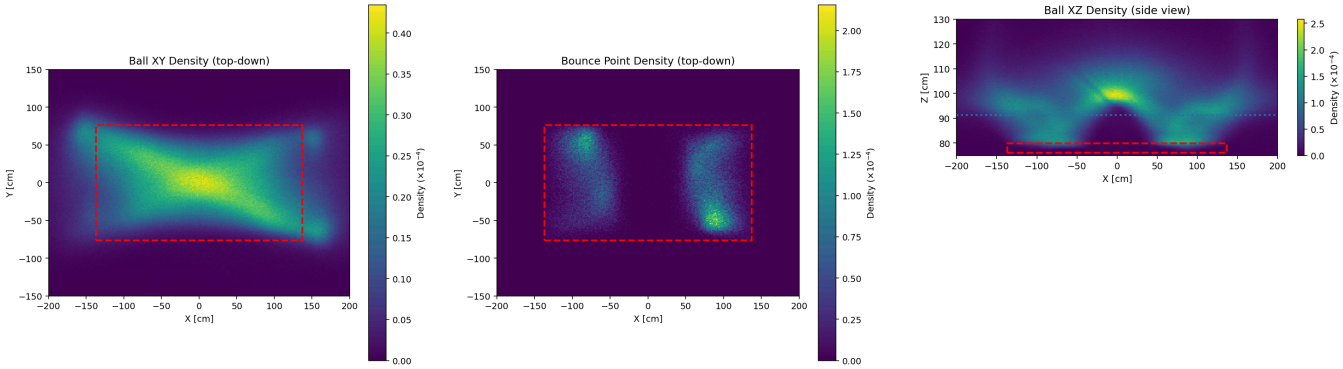


Figure 4 .

图4.

Ball position densities for the TT4D dataset.

TT4D 数据集的球位置密度。

The table region is marked by the dashed red line, and the net’ s height is marked by the dotted blue line.

球台区域由红色虚线标记，球网的高度由蓝色虚线标记。

Three heatmaps displaying ball position and bounce densities for the TT4D dataset.

显示 TT4D 数据集的球位置和弹跳密度的三个热图。

A dashed red rectangle outlines the table boundaries in each plot.

红色虚线矩形勾勒出每个图中的球台边界。

The top-left plot shows a top-down view of ball XY density, revealing an X-shaped high-density pattern that is concentrated in the center over the net and spreads towards the corners.

左上图显示了球 XY 密度的自上而下视图，显示了 X 形高密度图案，该图案集中在球网上方的中心并向角落扩散。

The top-right plot shows top-down bounce point density, featuring concentrated clusters of bounces on the left and right halves of the table.

右上图显示了自上而下的弹跳点密度，其中弹跳集中在球台的左半部分和右半部分。

The bottom plot presents a side view of ball XZ density, showing the ball's typical flight path arching closely over a dotted blue line that represents the net height, and dipping down to contact the table surface on both sides.

下图为球位置 XZ 密度的侧视图：典型球路以较低弧线越过表示网高的蓝色虚线，并在球网两侧下降触台。

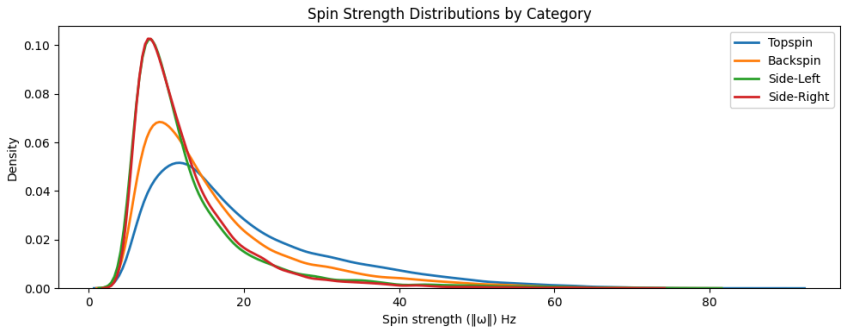


Figure 5 .

图5.

Visualization of ball spin strength per spin category.

每个旋转类别的球旋转强度的可视化。

A density plot visualizing ball spin strength distributions categorized into Topspin (blue), Backspin (orange), Side-Left (green), and Side-Right (red).

密度图可视化球旋转强度分布，分为上旋（蓝色）、下旋（橙色）、左侧（绿色）和右侧（红色）。

The horizontal axis measures Spin strength in Hz, and the vertical axis measures Density.

水平轴测量旋转强度（以 Hz 为单位），垂直轴测量密度。

The Side-Left and Side-Right curves are tightly aligned, displaying a single, sharp peak between 5 and 10 Hz before tapering off quickly.

左侧和右侧曲线紧密对齐，在 5 至 10 Hz 之间显示单个尖锐峰值，然后快速逐渐减小。

In contrast, the Topspin and Backspin curves are significantly flatter and exhibit heavy-tailed distributions; they feature an initial peak below 10 Hz, followed by a secondary, broader plateau that extends up to approximately 30 Hz.

相比之下，上旋和下旋曲线明显平坦，并呈现重尾分布；它们的初始峰值低于 10 Hz，随后是一个次要的、更宽的平台，延伸至大约 30 Hz。

The ball position densities in Figure 4 demonstrate interesting aspects of competitive table tennis gameplay.

图 4 中的球位密度展示了竞技乒乓球比赛的有趣方面。

The ball typically crosses between 5-15 cm above the net, as players keep the ball low to achieve high speed shots.

球通常在球网上方 5 至 15 厘米之间穿过，球员将球保持在较低位置以实现高速击球。

Players often choose cross-court shots, which are safer to land at higher speeds, over down-the-line hits.

相比直线球，球员常选择斜线球，因为在较高球速下，斜线球更容易安全落台。

The bounce point density also reveals insightful patterns.

弹跳点密度也揭示了有洞察力的模式。

When players send the ball from the left to the right in a cross-court hit, they tend to hit a clear high-density region.

球员从左向右打斜线球时，落点通常集中在一个明显的高密度区域。

However, when players hit the ball from the right to the left, the bounce-point density is far less concentrated.

然而，当球员从右向左击球时，弹跳点密度就远不那么集中。

We further analyze gameplay patterns by examining the distribution of spin strengths in Figure 5 .

我们通过检查图 5 中旋转强度的分布来进一步分析游戏模式。

For each segment, we extract the first predicted spin vector and assign it to one of five categories: “topspin", “backspin", “sidespin-left", “sidespin-right", and “no spin".

对于每个片段，我们提取第一个预测的旋转向量并将其分配给五个类别之一：“上旋”、“下旋”、“左旋”、“右旋”和“无旋转”。

A spin vector ω is classified as “no spin" when $\|\omega\|_\infty \leq 5$ Hz ; all remaining vectors are categorized following the conventions of (Kienzle et al., 2025 ; Kienzle et al., 2026) .

当 $\|\omega\|_\infty \leq 5$ Hz 时，旋转向量 ω 被分类为“无旋转”；所有剩余向量均按照 (Kienzle et al., 2025; Kienzle et al., 2026) 的约定进行分类。

The resulting distributions reveal differences between spin types.

所得的分布揭示了旋转类型之间的差异。

While all spin types exhibit a unimodal distribution, the “topspin" and “backspin" display slightly heavier-tailed distributions.

虽然所有旋转类型都表现出单峰分布，但“上旋”和“下旋”显示出稍微重尾的分布。

These patterns highlight the broader variability and more extreme spin magnitudes characteristic of top- and backspin strokes in competitive rallies.

这些模式凸显了竞技回合中上旋和下旋击球的更广泛的可变性和更极端的旋转幅度特征。

5. Evaluation of Lifting Network

5. 三维提升网络评估

Table 2 .

表 2.

Evaluation of the Lifting Network on the synthetic and TTST (Kienzle et al., 2025) datasets under augmentations. ✓/× indicates the presence/absence of augmentation.

在增强下的合成数据集和 TTST（Kienzle 等人，2025）数据集上评估三维提升网络。 ✓/× 表示存在/不存在增强。

The model shows strong resilience to lower frame rates and missing data.

模型对降低帧率和检测缺失表现出较强鲁棒性。

Augmentation 增强		Synthetic Dataset 合成数据集		TTST Dataset (Real) TTST 数据集（真实）	
Half FPS 半帧率	Missing Detections 缺失检测	$\Delta\vec{r}_{3D}$ (cm) ↓ (3D Pos. Error) $\Delta\vec{r}_{3D}$（cm）↓（3D位置误差）	$\Delta\vec{\omega}$ (Hz) ↓ (Spin Error) $\Delta\vec{\omega}$（Hz）↓（旋转误差）	$\Delta\vec{r}_{2D}$ (px) ↓ (2D Proj. Error) $\Delta\vec{r}_{2D}$（px）↓（二维投影误差）	Macro F1 ↑ (Spin Classif.) 宏F1 ↑（旋转分类）
×	×	2.60 ± 1.15	6.36 ± 2.76	2.52 ± 0.86	1.000
✓	×	3.28 ± 1.45	7.55 ± 3.41	2.34 ± 0.76	1.000
×	✓	2.69 ± 1.17	6.50 ± 2.81	2.78 ± 1.00	1.000
✓	✓	3.55 ± 1.63	7.86 ± 3.62	3.49 ± 2.44	0.941

Table 3 . 3D reconstruction error $\Delta\vec{r}_{3D}$ (cm) on TT4DBench.

表 3. TT4DBench 上的 3D 重建误差 $\Delta\vec{r}_{3D}$ (cm)。

Processing the full point (our approach) consistently yields lower errors than processing individual segments (traditional approach), demonstrating the benefit of full context.

一次处理完整回合（本文方法）始终比逐个处理独立片段（传统方法）误差更低，体现了完整上下文的价值。

		Full Point 完整回合		Individual Segments 独立击球片段	
Noise 噪音	View 视角	Mean 平均值	Std 标准差	Mean 平均值	Std 标准差
True 有	Back 后方视角	21.93	11.34	32.42	40.98
	Side 侧面	13.11	8.28	15.04	16.04
	Oblique 倾斜	18.71	6.86	21.43	16.56
	All Views 所有视图	17.92	9.73	22.96	28.08
False 无	Back 后方视角	20.68	10.88	30.14	39.10
	Side 侧面	12.81	8.07	14.65	16.41
	Oblique 倾斜	18.28	6.96	20.84	16.56
	All Views 所有视图	17.26	9.39	21.88	27.04

Table 4 .

表 4.

Side-view 3D reconstruction error $\Delta\vec{r}_{3D}$ (cm) for individual segments from TT4DBench.

TT4DBench 各个片段的侧视 3D 重建误差 $\Delta\vec{r}_{3D}$ (cm)。

Comparison to baseline methods with and without noisy 2D detections.

与有或没有噪声 2D 检测的基线方法的比较。

Noise 噪音	Method 方法	Individual Segments 独立击球片段 Mean 平均值	Std 标准差
True 有	TT3D TT3D	29.86	26.64
	LATTE-MV LATTE-MV	15.27	6.36
	TT4D (Ours) TT4D（我们的）	15.04	16.04
False 无	TT3D TT3D	29.91	27.79
	LATTE–MV LATTE-MV	15.78	6.37
	TT4D (Ours) TT4D（我们的）	14.65	16.41

We compare our Full-Sequence Lifting Network to traditional regression pipelines, evaluate the model' s robustness, show that the full-point setting is beneficial, and finally evaluate the physical plausibility of the predictions.

我们将全序列三维提升网络与传统回归流程进行比较，评估模型的鲁棒性，表明完整回合设置是有益的，最后评估预测的物理合理性。

Metrics.

指标。

For datasets with 3D ground truth (synthetic and TT3D), we report the 3D Trajectory Error ($\Delta\vec{r}_{3D}$), which is the mean Euclidean distance between the predicted and ground truth 3D positions, and F1 scores for table bounce and racket hit detection using a tolerance of ± 2 frames ($\pm 0.08\,s$).

对于具有 3D 真值（合成和 TT3D）的数据集，我们报告 3D 轨迹误差 ($\Delta\vec{r}_{3D}$)，它是预测和真值 3D 位置之间的平均欧几里得距离，以及使用 ± 2 帧容差 ($\pm 0.08\,s$) 的球台反弹和球拍击中检测的 F1 分数。

On synthetic data, we additionally measure the 3D Spin Error ($\Delta\vec{\omega}$), the mean Euclidean distance between predicted and ground truth spin vectors.

在合成数据上，我们还测量了 3D 旋转误差 ($\Delta\vec{\omega}$)，即预测旋转向量和真值旋转向量之间的平均欧几里得距离。

For 2D benchmarks (TTST & TTHQ) that lack 3D ground truth, we report the 2D Reprojection Error ($\Delta\vec{r}_{2D}$) in pixels and the Macro F1 score for topspin/backspin classification as described in (Kienzle et al., 2026) .

对于缺乏 3D 真值的 2D 基准（TTST 和 TTHQ），我们报告以像素为单位的 2D 重投影误差 ($\Delta\vec{r}_{2D}$) 以及上旋/下旋分类的宏 F1 分数，如 (Kienzle et al., 2026) 中所述。

Detailed mathematical definitions for all metrics are in the supplementary material.

所有指标的详细数学定义都在补充材料中。

Robustness Analysis.

稳健性分析。

Our network must be robust to real-world video artifacts, such as low frame rates or dropped ball detections from occlusions, which are not present in “perfect" evaluation datasets.

网络需要应对真实视频中的低帧率、遮挡导致的漏检等问题，而这些情况在理想评估数据中并不存在。

We therefore simulate “in-the-wild” conditions on the synthetic test set and the real-world TTST test set (following the evaluation protocol of (Kienzle et al., 2026)).

因此，我们在合成测试集和现实世界的 TTST 测试集上模拟“真实应用场景”条件（遵循（Kienzle 等人，2026）的评估协议）。

We implement the augmentations: Half FPS (dropping every second frame) and Missing Detections (randomly removing 10% of detections to simulate occlusions).

设置两种扰动：帧率减半（每隔一帧丢弃一帧），以及检测缺失（随机移除 10% 的检测以模拟遮挡）。

Results are shown in Table 2 .

结果如表2所示。

The model yields good performance on both datasets.

该模型在两个数据集上都产生了良好的性能。

Applying the augmentations individually does not lead to significant performance degradation.

单独应用增强不会导致性能显着下降。

Even though combining both augmentations leads to a noticeable performance drop, the results are still strong.

尽管结合这两种增强会导致性能明显下降，但结果仍然很强劲。

In total, the results verify our model’ s robustness to the “in-the-wild" conditions.

总的来说，结果验证了我们的模型对“真实应用场景”条件的稳健性。

Interpolation Performance.

插值性能。

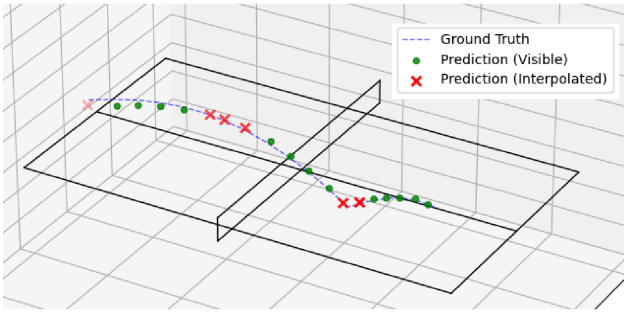


Figure 6 .

图6.

Reconstructed 3D trajectory.

重建的 3D 轨迹。

Even when the 2D detection is missing, the model is able to compute reasonable predictions due to its interpolation capabilities.

即使缺少 2D 检测，该模型也能够凭借其插值功能计算出合理的预测。

A 3D plot illustrating a reconstructed ball trajectory over a table tennis table.

3D 绘图展示了乒乓球桌上重建的球轨迹。

A legend indicates three distinct data representations: 'Ground Truth', 'Prediction (Visible)', and 'Prediction (Interpolated)'.

图例指示三种不同的数据表示形式：“真值”、“预测（可见）”和“预测（插值）”。

The plot demonstrates how the interpolated predictions successfully bridge the gaps in the trajectory where visible 2D detections are missing, closely following the expected path.

该图演示了插值预测如何成功弥合轨迹中缺少可见 2D 检测的间隙，紧密遵循预期路径。

Notably, Table 2 shows that the model yields strong performance under the Missing Detections augmentation, which randomly sets 10% of the ball detections to missed detections.

表 2 显示，模型在漏检增强条件下仍表现良好；这种增强会随机将 10% 的球检测结果设为漏检。

This not only shows the models robustness to real-world input, but emphasizes the model’s interpolation capabilities as the interpolated predictions only lead to minor performance degradations.

这不仅显示了模型对现实世界输入的鲁棒性，而且强调了模型的插值能力，因为插值预测只会导致轻微的性能下降。

This is further illustrated by Figure 6 which shows a reconstructed trajectory with successful interpolations.

图 6 进一步说明了这一点，该图显示了成功插值后的重建轨迹。

Full-Point vs. Single-Segment.

完整回合与单段。

We next validate our core contribution that processing the full, unsegmented sequence is superior to the “Traditional Pipeline” approach of processing isolated segments.

接下来，我们验证我们的核心贡献，即处理完整的、未分段的序列优于处理孤立片段的“传统流程”方法。

To create a challenging benchmark, we adapt the TT3D dataset (Gossard et al., 2025) , which provides ground-truth 3D trajectories.

为了创建具有挑战性的基准，我们采用了 TT3D 数据集（Gossard 等人，2025），它提供了真值 3D 轨迹。

While the original dataset is highly filtered and contains only single segments, we create a less-filtered stitched dataset (e.g. including throws) to obtain “in-the-wild” conditions.

原数据集经过严格筛选，只保留独立片段；我们采用筛选较少的数据并拼接片段，例如保留抛球过程，以构造接近真实应用的条件。

We denote this realistic multi-segment benchmark TT4DBench .

我们将这个现实的多段基准表示为 TT4DBench 。

We use this benchmark to compare our full-point approach against a single-segment approach.

我们使用这个基准来将我们的完整回合方法与单段方法进行比较。

In Table 3 we compare our model’s capability to process a full point at once vs processing each segment individually.

在表 3 中，我们比较了我们的模型一次处理完整回合的能力与单独处理每个片段的能力。

Processing the Full Point consistently outperforms the Individual Segments baseline across all camera views, reducing the mean $\Delta\vec{r}_{3D}$ from 21.88 cm to 17.26 cm.

在所有摄像机视图中，处理完整回合始终优于单个分段基线，将平均$\Delta\vec{r}_{3D}$从21.88厘米减少到17.26厘米。

This shows that the network successfully uses the extra context from the full point to improve reconstruction accuracy.

这表明网络成功地利用了完整回合的额外上下文来提高重建精度。

Comparison with Traditional Pipelines.

与传统流程的比较。

We benchmark our Full-Sequence Lifting Network in Table 4 against the methods of LATTE-MV (Etaat et al., 2025) and TT3D (Gossard et al., 2025) , which estimate the trajectory at inference time instead of using a learning based approach.

我们将表 4 中的全序列三维提升网络与 LATTE-MV (Etaat et al., 2025) 和 TT3D (Gossard et al., 2025) 的方法进行基准测试，这些方法在推理时估计轨迹，而不是使用基于学习的方法。

Since these methods are not able to reconstruct a full point, we restrict our evaluation to single segments.

由于这些方法无法重建完整回合，比较仅限于单个击球片段。

Since the LATTE-MV algorithm is limited to processing segments from a side-view perspective, we restrict its evaluation to side-view, single-segment examples.

由于 LATTE-MV 算法仅限于从侧视图角度处理片段，因此我们将其评估限制为侧视图、单片段示例。

Furthermore, LATTE-MV requires the precise 3D starting position and bounce point of the ball, information not available during in-the-wild inference.

此外，LATTE-MV 需要精确的 3D 起始位置和球的弹跳点，这些信息在真实应用场景推理期间无法获得。

To enable a comparison, we provide LATTE-MV with these values from the 3D ground truth, granting it a significant advantage with privileged 3D information.

为了进行比较，我们为 LATTE-MV 提供了 3D 真值中的这些值，使其在特权 3D 信息方面具有显着优势。

Despite this advantage, our method outperforms LATTE-MV and TT3D, achieving a lower mean reconstruction error.

尽管有这个优势，我们的方法优于 LATTE-MV 和 TT3D，实现了较低的平均重建误差。

Physical Consistency Check.

物理一致性检查。

Finally, we verify that our network’ s predictions are not just accurate but also physically consistent.

最后，我们验证网络的预测不仅准确，而且物理上一致。

After segmenting the point sequence in 3D, we analyze each individual segment.

在三维空间对回合序列分段后，我们逐段开展分析。

This per-segment validation is only possible because our Lift-First Pipeline provides robust time segmentations.

这种逐段验证之所以可行，是因为我们的 Lift-First Pipeline 提供了鲁棒的时间分段。

We fit a physics-based ODE model (Gossard et al., 2025) directly to our network’ s 3D output trajectories, acting as a proof of physical consistency.

我们将基于物理的 ODE 模型 (Gossard et al., 2025) 直接拟合到我们网络的 3D 输出轨迹，作为物理一致性的证明。

Note this differs from the original TT3D method, since we directly fit the ODE to our 3D predictions.

请注意，这与原始 TT3D 方法不同，因为我们直接将 ODE 拟合到我们的 3D 预测。

Figure 7 provides a clear qualitative validation: the fitted physical path (solid line) closely follows the network’ s predictions (dots), demonstrating the physical plausibility of our predictions.

图 7 提供了清晰的定性验证：拟合的物理路径（实线）紧密遵循网络的预测（点），证明了我们的预测的物理合理性。

This figure also visualizes the exact failure case of the “Traditional Pipeline": around the hit points, the 2D detections are missing (gaps in the dots) due to player occlusion.

该图还可视化了“传统流程”的确切失败案例：在击球点周围，由于球员遮挡而导致 2D 检测丢失（点中的间隙）。

Our network robustly predicts the 3D trajectory for the full sequence despite these gaps, yielding a continuous, physically plausible path where 2D-first methods would fail.

尽管存在这些间隙，我们的网络仍能稳健地预测整个序列的 3D 轨迹，从而产生一条连续的、物理上合理的路径，而 2D 优先的方法将失败。

As detailed in Section 3.6 , this ODE fit is also used as a quantitative quality metric during dataset curation.

如第 3.6 节所述，此 ODE 拟合还用作数据集管理期间的定量质量度量。

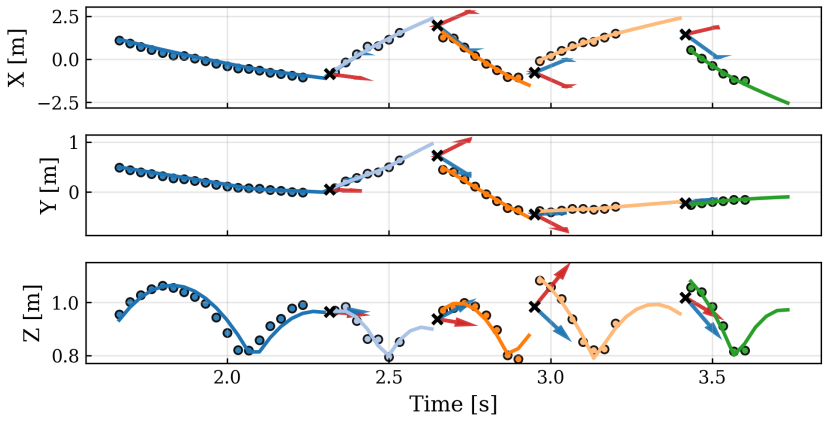


Figure 7 .

图 7.

Physical consistency: A physics-based ODE (solid line) fits the network’ s predicted 3D points (dots) with high precision, confirming the physical consistency of our output.

物理一致性：基于物理的 ODE（实线）高精度地拟合网络预测的 3D 点（点），从而确认我们输出的物理一致性。

Crosses are estimated racket hit points and the corresponding red and blue arrow are respectively the ball speeds before and after, used for the racket stroke estimation.

十字是估计的球拍击球点，对应的红色和蓝色箭头分别是前后的球速，用于球拍击球的估计。

A chart with three vertically stacked subplots displaying the X, Y, and Z coordinates of a ball's trajectory in meters against time in seconds.

具有三个垂直堆叠子图的图表，显示球轨迹的 X、Y 和 Z 坐标（以米为单位）与时间（以秒为单位）。

In each subplot, discrete dots representing the network's predicted 3D points are closely fitted by a continuous solid line representing the physics-based ODE.

在每个子图中，代表网络预测 3D 点的离散点与代表基于物理的 ODE 的连续实线紧密配合。

Black crosses indicate estimated racket hit points along the trajectory.

黑色十字表示沿轨迹估计的球拍击球点。

Originating from these crosses are red arrows, showing the ball's speed before the hit, and blue arrows, showing the ball's speed after the hit.

源于这些十字的是红色箭头，显示击球前的球速度，以及蓝色箭头，显示击球后的球速度。

Time Segmentation.

时间分段。

Using our detection method on the predicted 3D trajectories, we achieve F1-scores of 0.92 for table bounces and 0.83 for racket hits.

使用我们对预测 3D 轨迹的检测方法，我们的球台弹跳 F1 得分为 0.92，球拍击球得分为 0.83。

The lower racket hit performance is mainly due to limited ball visibility near the players, making accurate annotation difficult.

较低的球拍击球性能主要是由于球员附近的球能见度有限，导致准确标注变得困难。

Inference Speed.

推理速度。

At inference time, the model processes more than 500 points/s on a 10-year-old Titan X GPU, enabling reconstruction of thousands of games in minutes.

推理时，模型在已有约 10 年历史的 Titan X GPU 上每秒可处理超过 500 个回合，使几分钟内重建数千场比赛成为可能。

This contrasts with optimization-based methods such as LATTE-MV (Etaat et al., 2025) , TT3D (Gossard et al., 2025) , and MonoTrack (Liu and Wang, 2022) , which require a separate fitting procedure for each segment during inference.

这与基于优化的方法形成对比，例如 LATTE-MV (Etaat et al., 2025)、TT3D (Gossard et al., 2025) 和 MonoTrack (Liu and Wang, 2022)，这些方法在推理过程中需要为每个片段单独的拟合过程。

We present quantitative speed comparisons in the supplementary materials.

我们在补充材料中提供了定量的速度比较。

6. Applications

6. 应用

6.1. Generative Model of Competitive Gameplay

6.1.竞技比赛的生成模型

As a key downstream application, we utilize the fidelity and diversity of our dataset for training a generative model of competitive gameplay.

作为关键的下游应用程序，我们利用数据集的保真度和多样性来训练竞技比赛的生成模型。

High-quality, diverse generated samples serve as strong evidence that our TT4D dataset captures the underlying structure of the sport.

高质量、多样化的生成样本有力地证明我们的 TT4D 数据集捕获了这项运动的底层结构。

Generative Framework.

生成框架。

We adopt the Conditional Flow Matching (CFM) framework (Lipman et al., 2023 ; Tong et al., 2024a ; Tong et al., 2024b) .

我们采用条件流匹配（CFM）框架（Lipman et al., 2023; Tong et al., 2024a; Tong et al., 2024b）。

Let $\mathcal{D}$ be our dataset of reconstructed trajectories, $p_0 = \mathcal{N}(0, I)$ be a base distribution, and p_{data} be the empirical distribution over $\mathcal{D}$.

令$\mathcal{D}$为我们的重建轨迹数据集，$p_0 = \mathcal{N}(0, I)$为基础分布，p_{data}为$\mathcal{D}$上的经验分布。

Let τ denote a horizon of 20 observations, and c denote a history of 10 observations.

令 τ 表示含 20 个观测时刻的预测窗口，c 表示含 10 个观测时刻的历史窗口。

We learn a conditional vector field v_θ whose ODE solution $\frac{d\tau_t}{dt} = v_\theta(\tau_t, t|c)$ transports p_0 onto p_{data} .

我们学习一个条件向量场v_θ，其ODE解$\frac{d\tau_t}{dt} = v_\theta(\tau_t, t|c)$将$p_0$传输到$p_{\text{data}}$。

We use the flow-matching loss $\mathcal{L}_{\text{CFM}} = \mathbb{E}\left[\left\| v_\theta(\tau_t, t|c) - u_t \right\|_2^2\right]$, where τ_t is a simple interpolation between samples from p_0 and p_{data} , and u_t is the target velocity.

我们使用流匹配损失$\mathcal{L}_{\text{CFM}} = \mathbb{E}\left[\left\| v_\theta(\tau_t, t|c) - u_t \right\|_2^2\right]$，其中$\tau_t$是$p_0$和$p_{\text{data}}$样本之间的简单插值，$u_t$是目标速度。

This conditional formulation enables the model to generate full 20-step future trajectories with high physical fidelity given past context.

这种条件公式使模型能够在给定过去背景的情况下生成具有高物理保真度的完整 20 步未来轨迹。

Full details on the model architecture and training hyperparameters are in the supplementary material.

有关模型架构和训练超参数的完整详细信息位于补充材料中。

Evaluation.

评估。

We generate 10,000 multi-segment rallies by autoregressively rolling out the model: starting from a 10-observation history from our dataset, we generate the next 20 observations and then repeatedly invoke the flow model on the 10 most recent generated observations.

以数据集中的 10 个历史观测为起点，生成未来 20 个观测；随后反复将最近生成的 10 个观测输入流模型，自回归地生成 10,000 个包含多拍击球的回合。

These rallies are clipped using the same 3D-based time segmentation scheme outlined in Section 3.5 .

这些回合使用第 3.5 节中概述的相同的基于 3D 的时间分段方案进行剪辑。

Only 6 out of the 10,000 generated rallies failed the time segmentation stage.

在生成的 10,000 个回合中，只有 6 个未能通过时间分段阶段。

We evaluate the generated rallies using the same rigorous filtering and diversity metrics which we used to create the dataset.

我们使用与创建数据集相同的严格过滤和多样性指标来评估生成的回合。

As shown in Figure 8 , the distribution of the Physics-Based ODE Fit error for our generated samples (Gen Mean: 8.72 cm) is remarkably similar to and even slightly better than that of the real data (Data Mean: 10.77 cm).

如图 8 所示，我们生成的样本的基于物理的 ODE 拟合误差的分布（Gen 平均值：8.72 cm）与真实数据的分布（数据平均值：10.77 cm）非常相似，甚至略好一些。

This confirms that our generative model, trained on TT4D, produces physically plausible trajectories.

这证实了我们在 TT4D 上训练的生成模型可以产生物理上合理的轨迹。

We also demonstrate proper alignment in inter-hit time distribution.

生成数据与真实数据的相邻击球时间间隔分布也较为一致。

Though the generated gameplay distribution exhibits less spread, it nonetheless exhibits speeds at both ends of the real-world spectrum.

尽管生成的比赛过程分布范围较小，但它仍然在现实世界范围的两端表现出速度。

Hence, we should expect a broad class of behaviors in our dataset.

因此，我们应该期待数据集中出现广泛的行为。

We provide qualitative visualizations in the supplementary material, which show coherent long-horizon behavior.

我们在补充材料中提供定性可视化，显示出连贯的长期行为。

6.2. Racket Reconstruction

6.2.球拍重建

Our high-fidelity 3D ball trajectories and dense spin vectors provide a robust foundation for reconstructing racket motion.

我们的高保真 3D 球轨迹和密集旋转矢量为重建球拍运动提供了坚实的基础。

Since direct monocular tracking of the racket is often unreliable due to its high velocity, small size, and frequent occlusions, we leverage the reconstructed ball state instead.

由于球拍的高速度、小尺寸和频繁遮挡，直接单目跟踪通常不可靠，因此我们利用重建的球状态来代替。

We obtain the racket stroke parameters, defined as the racket’ s velocity and orientation at impact, from solving an inverse control problem.

我们通过求解逆控制问题获得球拍击球参数，定义为球拍撞击时的速度和方向。

By optimizing these stroke parameters such that the simulated post-impact ball trajectory accurately matches our observed 3D trajectory and bounce timing, we can recover the racket stroke that produced the observed ball trajectory.

通过优化这些击球参数，使模拟的击球后球轨迹与我们观察到的 3D 轨迹和弹跳时间精确匹配，我们可以恢复产生观察到的球轨迹的球拍击球。

We evaluate our approach against motion capture ground truth using infrared markers on the racket, across 92 recorded strokes.

我们使用球拍上的红外标记在 92 个记录的击球中评估我们针对运动捕捉真值的方法。

We obtain a mean orientation error of $26.4 \pm 4.4^{\circ}$ and a velocity error of 0.58 ± 0.40 m/s in the world frame, for an average impact speed of 3.72 m/s.

我们在世界坐标系中获得了$26.4 \pm 4.4^{\circ}$的平均方向误差和0.58 ± 0.40m/s的速度误差，平均撞击速度为 3.72 m/s。

The residual error is mainly in the world-frame Z-velocity and rotation about the X-axis (racket open/closed angle), likely due to unknown racket bounce properties.

剩余误差主要集中于世界坐标系的 Z 向速度及绕 X 轴的旋转（拍面打开或前倾的角度），可能与未知的球拍反弹特性有关。

6.3. Other Applications

6.3.其他应用

We believe this dataset unlocks significant potential for sports analytics , enabling advanced match visualization, human behavior analysis (Muelling et al., 2014) , and new training paradigms.

我们相信这个数据集释放了体育分析的巨大潜力，实现了高级比赛可视化、人类行为分析（Muelling et al., 2014）和新的训练范例。

For example, a ball launcher (Dittrich et al., 2023) could be conditioned to reproduce an opponent’ s specific playing style, offering individualized preparation for professional players.

例如，发球器（Dittrich et al., 2023）可以调节以重现对手的特定比赛风格，为职业球员提供个性化的准备。

The dataset also opens new frontiers in robot learning , e.g. providing the foundation for imitation learning, where a humanoid robot could learn to emulate professional-level strokes and rallies (Su et al., 2025 ; Zhang et al., 2026) .

该数据集也为机器人学习提供了新的机会，例如用于模仿学习，使人形机器人学习专业水平的击球动作和回合行为（Su 等，2025；Zhang 等，2026）。

We demonstrate this by training a motion tracking policy (Liao et al., 2025) on a retargeted motion (Araujo et al., 2026) from our dataset.

我们通过在数据集中的重定向运动（Araujo 等人，2026）上训练运动跟踪策略（Liao 等人，2025）来证明这一点。

Implementation details and results may be found in the supplementary material.

实施细节和结果可在补充材料中找到。

Furthermore, TT4D provides the necessary data for training behavior-aware models, such as anticipatory robotic policies, allowing agents to reason about the strategic decisions made by other players.

此外，TT4D 可为具有预判能力的机器人策略等行为感知模型提供训练数据，使智能体能够推断其他球员的策略决策。

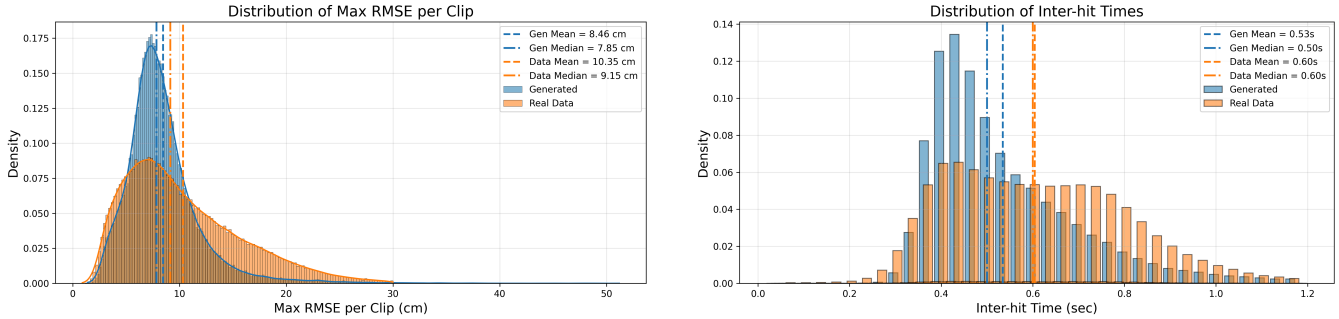


Figure 8 .

图 8.

(Top) Physical plausibility: The distribution of maximum ODE fit errors (RMSE) for generated rallies (blue) closely matches the real gameplay distribution (orange), indicating that our generative model produces physically valid 3D trajectories.

（上）物理合理性：生成的回合（蓝色）的最大 ODE 拟合误差 (RMSE) 分布与真实游戏分布（橙色）密切匹配，表明我们的生成模型生成物理上有效的 3D 轨迹。

(Bottom) Temporal realism: The inter-hit time distribution of generated rallies (blue) aligns closely with real gameplay (orange), demonstrating the model captures tempo and exchange dynamics.

下：时间真实性。生成回合（蓝色）的相邻击球时间间隔分布与真实比赛（橙色）接近，表明模型学到了比赛节奏及往返击球规律。

Two vertically stacked histograms comparing generated data in blue and real data in orange.

两个垂直堆叠的直方图，比较蓝色的生成数据和橙色的实际数据。

The top histogram displays the distribution of maximum ODE fit errors (RMSE), showing both distributions heavily overlapping and peaking near the lower end of the axis.

顶部直方图显示最大 ODE 拟合误差 (RMSE) 的分布，显示两个分布严重重叠并在轴下端附近达到峰值。

The bottom histogram displays the distribution of inter-hit times in seconds, showing both distributions overlapping with a primary peak between 0.4 and 0.6 seconds and sharing a similar right-skewed tail.

底部直方图显示了以秒为单位的击中时间分布，显示两个分布与 0.4 到 0.6 秒之间的主峰重叠，并共享类似的右偏尾部。

7. Conclusion

7. 结论

We presented the Lift-First Pipeline for multimodal sports reconstruction from monocular videos, which lifts entire unsegmented rallies to 3D before performing any time segmentation.

我们提出了用于从单目视频进行多模态运动重建的提升优先流程，该流程在执行任何时间分割之前将整个未分段的回合提升到 3D。

By working in the 3D domain, our approach avoids the fragility of 2D-based time segmentation under occlusions and missing detections.

通过在 3D 域中工作，我们的方法避免了基于 2D 的时间分割在遮挡和缺失检测下的脆弱性。

This is enabled by a Full-Sequence Lifting Network trained on a large-scale synthetic dataset of 3 M full points that also models the pre-serve toss.

这是通过全序列三维提升网络实现的，该网络在3M个完整回合的大规模合成数据集上进行训练，该数据集也对发球前投掷进行了建模。

While demonstrated on table tennis, our general pipeline can be readily extended to other sports such as tennis or badminton.

虽然在乒乓球上进行了演示，但我们的通用流程可以轻松扩展到其他运动，例如网球或羽毛球。

The result of this pipeline is the TT4D dataset , a 140+ hour high-fidelity multimodal dataset with 3D ball trajectories, 3D human meshes, and, for the first time at this scale, 3D spin vectors and precise 3D-derived time segmentations.

该流程的结果是 TT4D 数据集，这是一个 140 多个小时的高保真多模态数据集，具有 3D 球轨迹、3D 人体网格，并且首次在这种规模上提供 3D 旋转向量和精确的 3D 派生时间分段。

We demonstrated the fidelity and utility of TT4D through physics-based validation and downstream applications, including racket-contact estimation and a generative model of competitive rallies.

我们通过基于物理的验证和下游应用（包括球拍接触估计和竞争性回合生成模型）展示了 TT4D 的保真度和实用性。

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We provide additional details on our dataset generation and evaluation pipeline in this supplementary material.

我们在本补充材料中提供了有关数据集生成和评估流程的更多详细信息。

Moreover, additional experiments are evaluated and some qualitative examples are analyzed.

此外，还评估了其他实验并分析了一些定性示例。

In addition, we also illustrate one example rally in the dataset (Figure SM11) as well as one example rally generated by our generative algorithm (Figure SM12).

此外，我们还说明了数据集中的一个示例回合（图 SM11）以及由我们的生成算法生成的一个示例回合（图 SM12）。

We publish our dataset upon acceptance.

我们在接受后发布我们的数据集。

Our gameplay visualizations are powered by Viser (Yi et al., 2025) .

我们的游戏可视化由 Viser 提供支持（Yi 等人，2025）。

Appendix A Data Preprocessing Details

附录 A 数据预处理详细信息

This section provides implementation details for the Data Acquisition and Preprocessing stage of our pipeline.

本节提供我们流程的数据采集和预处理阶段的实现细节。

A.1. Video Clipping and Deduplication

A.1.视频剪辑和重复数据删除

Video Clipping Our reconstruction pipeline starts with a set of multi-hour, uncut table tennis competition videos.

视频剪辑 我们的重建流程从一组长达数小时、未剪辑的乒乓球比赛视频开始。

This is processed into a set of reconstructed points by first progressing through two stages of clipping.

原始视频先经过两个阶段的裁剪，再进入逐回合重建。

The first stage clips the broadcast video at frames when the scoreboard advances.

第一阶段在记分牌前进时按帧剪辑转播视频。

We detect the change in score by first identifying the scoreboard using YOLO (Varghese and M., 2024) and then recovering its text using PaddleOCR (Cui et al., 2025) .

我们首先使用 YOLO 识别记分板（Varghese 和 M.，2024），然后使用 PaddleOCR 恢复其文本（Cui 等人，2025）来检测分数的变化。

The text associated with the player names also allows us to discern singles from doubles gameplay.

球员姓名对应的文字也可用于区分单打与双打比赛。

This scoreboard-based clip contains a significant number of frames preceding and following the actual gameplay.

这个基于记分板的剪辑包含实际游戏之前和之后的大量帧。

To trim the clips down further, the second stage identifies the approximate start and end of the point by using the 2D ball trajectory, $b_{2D}(n)$.

第二阶段根据二维球轨迹 $b_{2D}(n)$ 确定回合的大致起止时刻，进一步裁剪片段。

For this stage, we use the LATTE-MV ball tracker (Etaat et al., 2025) to obtain $b_{2D}(n)$.

对于这个阶段，我们使用 LATTE-MV 球跟踪器 (Etaat et al., 2025) 来获得$b_{2D}(n)$。

This ball tracker uses the full TracknetV3 model, which includes the inpainting module.

该球跟踪器使用完整的 TracknetV3 模型，其中包括轨迹补全模块。

Note that the inpainting module is removed when we compute $b_{2D}(n)$ for reconstruction.

请注意，当我们计算$b_{2D}(n)$进行重建时，轨迹补全模块被删除。

The key idea of this stage is that while the oscillations of $b_{2D}(n)$ cannot reliably solve hit-point identification, it can be used to determine the approximate start and end of the point.

其关键在于：$b_{2D}(n)$ 的往返变化虽不足以可靠定位每次击球，却能用于确定整个回合的大致起止范围。

This procedure is outlined in Algorithm 1 .

算法 1 概述了该过程。

Note that this stage may fail and produce clips that do not exhibit any gameplay.

请注意，此阶段可能会失败并生成不展示任何比赛过程的剪辑。

This is not a problem, however, since these clips are removed in the filtering stage.

然而，这不是问题，因为这些剪辑在过滤阶段被移除。

Algorithm 1 Second Clipping Stage and Helper Functions

算法 1 第二个剪辑阶段和辅助函数

Helper Functions.

辅助功能。

ComputeConstantIntervals $(Signal, strength_th, time_th)$: Inputs: A 1D signal; a magnitude threshold defining "near-zero" values; a minimum interval duration.

ComputeConstantIntervals $(Signal, strength_th, time_th)$: 输入：一维信号；定义"接近零"值的幅度阈值；最小间隔持续时间。

Returns: All contiguous intervals where the signal remains below the threshold for longer than the required duration.

返回：信号保持低于阈值的时间超过所需持续时间的所有连续间隔。

FindOverlappingIntervals $(Intervals_1, Intervals_2)$: Inputs: Two sorted lists of disjoint intervals.

FindOverlappingIntervals $(Intervals_1, Intervals_2)$: 输入：两个不相交间隔的排序列表。

Returns: All sub-intervals formed by intersections of the two lists.

返回： 两个列表交集形成的所有子区间。

FindComplementaryIntervals $(Signal, Intervals, min_length)$: Inputs: A signal; a list of covered intervals; a minimum acceptable gap length.

FindComplementaryIntervals $(Signal, Intervals, min_length)$: 输入：一个信号；覆盖区间列表；可接受的最小间隙长度。

Returns: All uncovered intervals (gaps) of at least the

BestPointEstimate $(corr_x, Intervals, min_length, peak_th)$: Inputs: A correlation signal; candidate gameplay intervals; a minimum interval length; a peak magnitude threshold.

BestPointEstimate $(corr_x, Intervals, min_length, peak_th)$: 输入：相关信号；候选游戏间隔；最小间隔长度；峰值幅度阈值。

Returns: The index of the interval with the highest

Main Procedure: Second Clipping Stage

主要程序：第二剪切阶段

minimum length.

返回：所有至少具有最小长度的未覆盖间隔（间隙）。

number of above-threshold peaks (or -1 if none qualify).

返回： 具有最高数量的阈值以上峰值的区间索引（如果没有符合条件则返回-1）。

1: function FindApproximateStartEnd(x, y)

2: $horizontal_kernel \leftarrow [\times 15, \times 6, -\times 15]$

3: $vertical_kernel \leftarrow -[\times 20, -\times 20]$

4: $I_x \leftarrow \text{ComputeConstantIntervals}(x, 1, 20)$

5: $I_y \leftarrow \text{ComputeConstantIntervals}(y, 1, 20)$

6: $Zero \leftarrow \text{FindOverlappingIntervals}(I_x, I_y)$

7:

$Possible \leftarrow \text{FindComplementaryIntervals}(x, Zero, 80)$

8: $\delta x[t] \leftarrow x[t+1] - x[t]$

9: $\delta y[t] \leftarrow y[t+1] - y[t]$

10: $corr_x \leftarrow \text{Convolve}(\delta x, \text{reverse}(horizontal_kernel))$

11: $corr_y \leftarrow \text{Convolve}(\delta y, \text{reverse}(vertical_kernel))$

12: if $|corr_x| \neq |corr_y|$ then

13: return $(-1, -1)$

14: end if

15: Construct Boolean mask over

构造布尔掩码

$[0, |corr_x|)$ based on

基于

$Possible$

16: Zero out entries of

将以下条目归零

$corr_x$ and $corr_y$ where mask is False

其中 mask 为 False

17:

$Signal \leftarrow \text{FindComplementaryIntervals}(corr_x, Zero, 0)$

18: for each (s, e) in $Signal$ do

19: if $|(e - s) - 50| < 20$ then

▷ 1. Kernels and thresholds

1. 内核和阈值

▷ 2. Zero-valued intervals in both x and y

2. 两者的零值区间

▷ 3. Candidate gameplay intervals

3. 候选比赛时间区间

▷ 5. Mask correlations outside candidate intervals

5. 屏蔽候选区间之外的相关性

▷ 6. Extract correlation-domain intervals

6. 提取相关域区间

$dot \leftarrow \sum_{t=s}^{e} corr_x[t]\, corr_y[t]$

if $|dot| \geq 30000$ then

Zero out

归零

$corr_x,\ corr_y$ on $[s, e]$

end if

end if

end for ▷ 7. Choose best gameplay interval

7. 选择最佳比赛时间区间

$idx \leftarrow \text{BestPointEstimate}(corr_x, Signal, 60, 50)$

if $idx = -1$ then

return $(-1, -1)$

end if

$(start, end) \leftarrow Signal[idx]$ ▷ 8. Reject clips where the ball remains still too long

$dx \leftarrow \Delta x_S,\ dy \leftarrow \Delta y_S$ 8. 拒绝球静止时间过长的片段

if $|\{t : dx[t] = 0,\ x[t] \neq 0,\ dy[t] = 0,\ y[t] \neq 0\}| > 30$ then

return $(-1, -1)$

end if

return $(start, end)$

end function

Duplicated Frame Removal While processing online table-tennis footage, we observed that certain frames were duplicated within the video stream.

重复帧删除在处理在线乒乓球镜头时，我们观察到某些帧在视频流中重复。

This phenomenon typically arises when the frame rate used for camera capture differs from that used during rendering or re-encoding.

当用于相机捕获的帧速率与渲染或重新编码期间使用的帧速率不同时，通常会出现这种现象。

In such cases, the renderer/encoder fills missing timestamps by repeating the temporally nearest frames, thereby matching the target output frame rate.

在这种情况下，渲染器/编码器通过重复时间上最近的帧来填充丢失的时间戳，从而匹配目标输出帧速率。

We identified three cases: (i) periodic duplicated frames, characterized by a first duplicated index s and a duplication period f, (ii) aperiodic duplicated frames, and (iii) no duplicated frames.

我们确定了三种情况：(i) 周期性重复帧，其特征是第一个重复索引s和重复周期f，(ii) 非周期性重复帧，以及 (iii) 无重复帧。

The periodic case was the most common in the processed videos.

在处理的视频中，周期性情况是最常见的。

Although imperceptible to the human eye, this frame-duplication artifact causes the tracked ball to appear at the same position for two consecutive frames.

虽然人眼无法察觉，但这种帧重复伪影会导致跟踪球在连续两个帧中出现在同一位置。

As a consequence, the reconstruction algorithm falsely interprets these repetitions as sudden deceleration or temporary stops, which deteriorates trajectory estimation accuracy.

因此，重建算法错误地将这些重复解释为突然减速或暂时停止，这降低了轨迹估计的准确性。

A direct pixel-wise absolute difference is not a reliable measure of frame similarity.

直接的像素级绝对差并不是帧相似性的可靠度量。

Even when two frames are visually identical, their difference image resembles approximately white noise due to encoder and compression artifacts.

即使两帧视觉上相同，由于编码器和压缩伪影，它们的差异图像也近似白噪声。

The aggregated absolute difference over all pixels in such cases can be comparable to that between genuinely different consecutive frames.

此时，所有像素绝对差的总和可能与真正不同的相邻帧相当，因而难以区分两者。

This motivates the use of SSIM , which evaluates perceptual similarity by comparing luminance, contrast, and structural content.

这激发了 SSIM 的使用，它通过比较亮度、对比度和结构内容来评估感知相似性。

Using a high SSIM threshold allows us to detect and discard duplicated frames, though occasional false positives and false negatives may still occur.

使用高 SSIM 阈值允许我们检测并丢弃重复的帧，尽管偶尔仍可能出现误报和漏报。

To robustly detect periodic duplication, let $\{u_i\}_{i=1}^{N}$ denote the sorted indices of duplicated frames.

为了稳健地检测周期性重复，让$\{u_i\}_{i=1}^{N}$表示重复帧的排序索引。

We consider the inter-duplicate spacings $\Delta_i = u_{i+1} - u_i, \text{for} i = 1, ..., N-1,$ and ignore trivial repetitions with $\Delta_i = 1$.

我们考虑重复项之间的间距$\Delta_i = u_{i+1} - u_i, \text{for} i = 1, ..., N-1,$并忽略$\Delta_i = 1$的琐碎重复。

Throughout this paper, we use $|\mathcal{A}|$ to denote the cardinality of a set $\mathcal{A}$.

本文以 $|\mathcal{A}|$ 表示集合 $\mathcal{A}$ 的基数，即元素个数。

We estimate the duplication period as the modal spacing

我们取重复帧间距的众数作为重复周期的估计：

(SM1)
$$\hat{f} = \underset{d \in \mathbb{N}}{\operatorname{argmax}} \left| \{ i \mid \Delta_i = d,\ \Delta_i > 1 \} \right| .$$

Assuming periodic duplicated frames follow the model $u_k = s + kf$, the start offset is obtained as the most frequent residue modulo $\hat{f}$:

假设周期性重复帧满足模型 $u_k = s + kf$，则对帧索引取模 $\hat{f}$，以出现最多的余数作为起始偏移量：

(SM2)
$$\hat{s} = \arg\max_{r \in \{0, ..., \hat{f}-1\}} \sum_{i=1}^{N} \left(u_i \equiv r \pmod{\hat{f}} \right),$$

where $\mathbf{1}(\cdot)$ denotes the indicator function.

其中$\mathbf{1}(\cdot)$表示指示函数。

If fewer than three duplicated indices exist, or if no non-trivial spacings are observed, periodicity cannot be estimated reliably and the case is treated as aperiodic.

如果存在少于三个重复索引，或者没有观察到不平凡的间距，则无法可靠地估计周期性，并且该情况被视为非周期性的。

Finally, the video is re-encoded with the corrected frame rate after removing all detected duplicated frames.

最后，在删除所有检测到的重复帧后，使用校正后的帧速率对视频进行重新编码。

A.2. Camera Calibration

A.2.相机标定

The camera parameters are not provided for online videos, so the calibration must be performed without a dedicated calibration pattern.

在线视频不提供相机参数，因此必须在没有专用校准图案的情况下进行校准。

We therefore adopt the TT3D approach (Gossard et al., 2025) , which uses the table itself as the calibration object.

因此，我们采用 TT3D 方法（Gossard et al., 2025），该方法使用球台本身作为校准对象。

The world frame is defined by the standard table geometry, and the four table corners serve as known 3D reference points.

世界坐标系由标准球台几何形状定义，四个球台角用作已知的 3D 参考点。

A segmentation model first extracts the table mask, after which table edges are detected using a Hough line transform and intersected to obtain the corner locations.

分割模型首先提取球台掩膜，然后使用霍夫线变换检测球台边缘并相交以获得角位置。

With these correspondences, the camera extrinsics, rotation $\mathbf{R}$ and translation $\mathbf{T}$ relative to the table frame, are estimated jointly with the focal length f .

利用这些对应关系，联合估计相机相对于球台坐标系的旋转 $\mathbf{R}$、平移 $\mathbf{T}$ 及焦距 f。

The problem is formulated as a Perspective- n -Point task with unknown focal length and solved by iteratively minimizing the reprojection error over $(\mathbf{R}, \mathbf{T}, f)$, enabling robust calibration even under partial occlusions and in unconstrained broadcast footage.

该问题被表述为焦距未知的透视-n-点任务，并通过迭代最小化$(\mathbf{R}, \mathbf{T}, f)$上的重投影误差来解决，即使在部分遮挡和不受约束的广播镜头中也能实现稳健的校准。

In contrast to the original algorithm in (Gossard et al., 2025) , our segmentation model is based on the UNet++ architecture (Zhou et al., 2018) with the bigger Efficientnet-B0 backbone (Tan and Le, 2019) .

与（Gossard 等人，2025）中的原始算法相比，我们的分割模型基于 UNet++ 架构（Zhou 等人，2018），具有更大的 Efficientnet-B0 主干网（Tan 和 Le，2019）。

For training, we use a combination of Binary Cross Entropy and DICE Loss.

对于训练，我们结合使用二元交叉熵和 DICE 损失。

With these modifications, we are able to obtain high quality camera calibrations for our dataset.

经过上述修改，我们为数据集获得了高质量的相机标定结果。

A.3. Player Reconstruction

A.3.球员重建

We use 4DHumans (Goel et al., 2023) to track and reconstruct the human players.

我们使用 4DHumans (Goel et al., 2023) 来跟踪和重建人类球员。

The human body is represented using the SMPL model (Loper et al., 2015) , and the parameters are given in the camera frame.

人体使用 SMPL 模型（Loper 等，2015）表示，其参数最初定义在相机坐标系中。

We orient the SMPL mesh in the world frame using the camera rotation.

我们使用相机旋转在世界坐标系中定向 SMPL 网格。

To position the SMPL mesh in the world frame, we first use the camera extrinsics to compute the homography transform from the image plane to the ground plane.

为了在世界坐标系中定位 SMPL 网格，我们首先使用相机外参来计算从图像平面到地平面的单应性变换。

Next, we assume that the rotated 3D keypoint closest to the ground is on the ground plane, and we use the homography matrix to map the corresponding 2D keypoint to its 3D location on the ground plane.

接下来，我们假设最接近地面的旋转 3D 关键点位于地平面上，并使用单应矩阵将相应的 2D 关键点映射到其在地平面上的 3D 位置。

We verify global 3D consistency by checking that the average location of all reconstructed 3D human meshes is plausibly near the 3D table.

我们通过检查所有重建的 3D 人体网格的平均位置是否合理地靠近 三维球台来验证全局 3D 一致性。

This step is particularly important to filter non-players like referees or the crowd, common misdetections that are especially present for doubles gameplay.

此步骤对于过滤裁判或人群等非球员尤其重要，这是双打比赛中尤其存在的常见误检。

Appendix B Lifting Network Details

附录 B 三维提升网络详细信息

Algorithm 2 Stitching Algorithm

算法2 拼接算法

1: Input: Data pools

数据池

$P_{\text{throw}}, P_{\text{serve}}, P_{\text{return}}$

2: Output: A list of stitched trajectory segments

拼接轨迹段列表

$Point$

3:

4: function BuildStitchedPoint

5: $Point \leftarrow \text{EmptyList}()$

6:

7: $initial_conditions_throw \leftarrow P_{\text{throw}}.\text{Sample}()$ ▷ 1. Simulate the initial ball toss

1. 模拟初始抛球

8: $throw_seg \leftarrow \text{Simulate}(initial_conditions_throw)$

9: $r_{\text{start}} \leftarrow throw_seg.\text{end_positon}$

10: $Point.\text{Add}(throw_seg)$

11:

12: $serve_seg \leftarrow \text{StitchNextSegment}(r_{\text{start}}, P_{\text{serve}})$ ▷ 2. Stitch the serve segment

2. 拼接发球段

13: if $serve_seg$ is null then

14: return FailedPoint

15: end if

16: $Point.\text{Add}(serve_seg)$

17: $r_{\text{start}} \leftarrow serve_seg.\text{end_position}$

18:

19: for $i \leftarrow 1$ to MAX_Point_LENGTH do ▷ 3. Recursively stitch return segments

3. 递归拼接回球片段

20: $return_seg \leftarrow \text{StitchNextSegment}(r_{\text{start}}, P_{\text{return}})$

21: if $return_seg$ is null then

22: return FailedPoint

23: end if

24: $Point.\text{Add}(return_seg)$

25: $r_{\text{start}} \leftarrow return_seg.\text{end_position}$

26: end for

27:

```
    return Point
end function

function StitchNextSegment(r_start, Pool)                ▷ Multiple tries to find the next segment
    for i ← 1 to MAX_ATTEMPTS do                          多次尝试寻找下一段

        prior ← Pool.FindClosestPrior(r_start)            ▷ Use initial velocity and spin of close match
        v_new ← prior.velocity                            使用接近匹配的初始速度和旋转
        ω_new ← prior.spin
        trajectory ← Simulate(r_start, v_new, ω_new)

        if (IsValidTrajectory(trajectory)) then
            return trajectory                             ▷ Check if we simulated a valid table tennis trajectory
        end if                                            检查我们是否模拟了有效的乒乓球轨迹
    end for

    return null
end function
```

B.1. Network Architecture

B.1.网络架构

Our lifting network builds upon the 2D-to-3D lifting transformer introduced by Kienzle et al. (2026) .

我们的三维提升网络建立在 Kienzle 等人引入的 2D 到 3D 三维提升Transformer的基础上。（2026）。

Its core design takes a sequence of N 2D ball detections $\{\vec{r}_{2D}(t_n)\}_{n=0}^{N-1} \in \mathbb{R}^{N\times 2}$, the corresponding exact times $\{t_n\}_{n=0}^{N-1} \in \mathbb{R}^N$ that can be derived from the videos framerate, and a set of 13 predefined 2D table keypoints $\{\vec{k}_i\}_{i=0}^{12} \in \mathbb{R}^{13\times 2}$ as input.

其核心设计采用一系列N2D球检测$\{\vec{r}_{2D}(t_n)\}_{n=0}^{N-1} \in \mathbb{R}^{N\times 2}$、可以从视频帧速率导出的相应精确时间$\{t_n\}_{n=0}^{N-1} \in \mathbb{R}^N$以及一组13个预定义2D球台关键点$\{\vec{k}_i\}_{i=0}^{12} \in \mathbb{R}^{13\times 2}$作为输入。

These 13 keypoints implicitly provide the network with the full camera calibration information needed for trajectory lifting.

这 13 个关键点隐式地为网络提供了轨迹三维提升所需的完整相机标定信息。

The ball detections and 13 table keypoints are first embedded into sequence of N "location tokens" $\{l_n\}_{n=0}^{N-1} \in \mathbb{R}^{d_{\text{model}}}$ with embedding dimension d_{model}.

球检测和 13 个球台关键点首先嵌入到具有嵌入维度d_{model}的N“位置词元”$\{l_n\}_{n=0}^{N-1} \in \mathbb{R}^{d_{\text{model}}}$序列中。

We implement a custom Disentangled Context Embedding (DCE), that projects the ball positions and table keypoints separately into higher dimension vectors, keeping the ball information disentangled from the camera calibration information.

我们实现了自定义的解耦上下文嵌入（DCE），它将球位置和球台关键点分别投影到更高维度的向量中，使球信息与相机标定信息分离。

This now allows replacing a ball vector with a learnable interpolation token in case the ball was not correctly detected.

现在，如果未正确检测到球，则可以用可学习的插值词元替换球向量。

Finally, ball vector and table keypoints vector are concatenated and projected to compute the location token.

最后，将球向量和球台关键点向量连接并投影以计算位置词元。

This is visualized in fig.

这如图所示。

SM1 .

SM1。

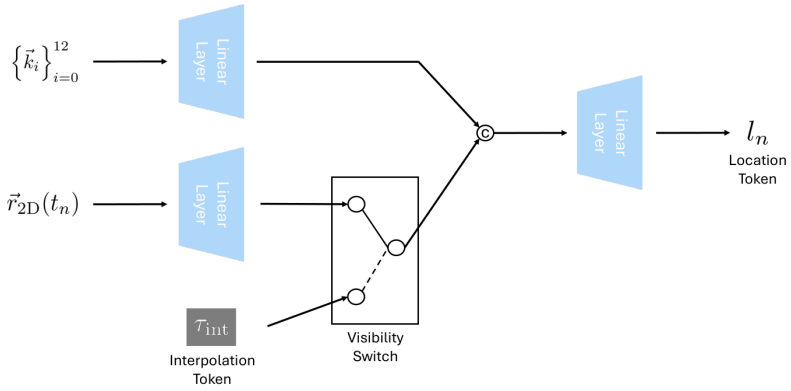


Figure SM1 .

图SM1。

Schematic illustration of the Disentangled Context Embedding (DCE).

解耦上下文嵌入 (DCE) 的示意图。

Table Keypoints $\{\vec{k}_i\}_{i=0}^{12}$ and detected 2D ball coordinates $\vec{r}_{2D}$ are projected to higher dimensional vectors.

球台关键点$\{\vec{k}_i\}_{i=0}^{12}$和检测到的二维球坐标$\vec{r}_{2D}$被投影到更高维度的向量。

If a ball is not correctly detected, a learnable interpolation token τ_{int} is used instead of the ball vector.

如果没有正确检测到球，则使用可学习的插值词元τ_{int}来代替球向量。

Finally, both vectors are concatenated and the final location token l_n is computed.

最后，连接两个向量并计算最终位置词元l_n。

A block diagram illustrating the Disentangled Context Embedding process.

解耦上下文嵌入过程的框图。

Two parallel input paths for Table Keypoints and detected 2D ball coordinates are shown being projected into higher dimensional vectors.

显示球台关键点和检测到的 2D 球坐标的两个并行输入路径被投影到更高维度的向量中。

A conditional routing step shows that a learnable interpolation token is substituted for the ball vector if the ball is not correctly detected.

条件路由步骤表明，如果未正确检测到球，则用可学习的插值词元替换球向量。

The diagram then shows these two resulting vectors being concatenated together to compute the final output, designated as the location token.

然后，该图显示这两个结果向量连接在一起以计算最终输出，指定为位置词元。

To effectively integrate these interpolation tokens into the transformer without degrading the feature quality of observed frames, we adopt the Deferred Upsampling Token Attention (DUTA)

mechanism (Einfalt et al., 2023) .

为了有效地将这些插值词元集成到Transformer中，而不降低观察到的帧的特征质量，我们采用了延迟上采样词元注意力（DUTA）机制（Einfalt et al., 2023）。

Since the learnable interpolation tokens initially contain no specific ball information beyond their temporal embedding, allowing valid location tokens to attend to them in the first layers can introduce significant noise and corrupt the high-fidelity representations of detected ball positions.

由于可学习插值词元最初不包含超出其时间嵌入的特定球信息，因此允许有效位置词元在第一层中处理它们可能会引入显着的噪声并破坏检测到的球位置的高保真表示。

DUTA addresses this by restricting the self-attention mask in the initial transformer layers: valid location tokens are prohibited from attending to interpolated location tokens, while interpolated location tokens are permitted to attend to valid location tokens.

DUTA 限制初始 Transformer 层的注意力掩膜：有效位置词元不能关注插值词元，而插值词元可以关注有效位置词元。

This ensures that the interpolated location tokens can effectively aggregate context from their neighbors to estimate missing positions without deteriorating the features of the rest of the sequence.

这确保了插值位置词元可以有效地聚合来自其邻居的上下文以估计丢失的位置，而不会恶化序列其余部分的特征。

Moreover, a key architectural extension is the removal of the learnable "spin token" $\mathbf{s}$ used in the baseline, as this cannot be extended for processing full points.

另一项关键修改是移除基线的可学习旋转词元 $\mathbf{s}$，因为该设计无法直接适配完整回合。

Instead of predicting a single initial spin vector, we modify the network to predict spin in a dense, per-frame manner by applying a small Spin Head, consisting of a small 3-layer MLP, to every output location token l_n .

我们不是预测单个初始旋转向量，而是修改网络，通过将由小型 3 层 MLP 组成的小型旋转头应用于每个输出位置词元l_n，以密集的每帧方式预测旋转。

Consequently, the network outputs a full sequence of dense spin vectors $\{\vec{\omega}(t_n)\}_{n=0}^{N-1} \in \mathbb{R}^{N\times 3}$ alongside the 3D trajectory sequence $\{\vec{r}_{3\mathrm{D}}(t_n)\}_{n=0}^{N-1} \in \mathbb{R}^{N\times 3}$.

因此，网络会输出密集旋转向量的完整序列$\{\vec{\omega}(t_n)\}_{n=0}^{N-1} \in \mathbb{R}^{N\times 3}$以及 3D 轨迹序列 $\{\vec{r}_{3\mathrm{D}}(t_n)\}_{n=0}^{N-1} \in \mathbb{R}^{N\times 3}$。

This dense prediction provides a much richer output than prior methods and resolves the ambiguity of defining an "initial" spin for an unsegmented sequence.

这种密集预测提供了比先前方法更丰富的输出，并解决了为未分段序列定义“初始”旋转的模糊性。

A schematic overview of our modified lifting network is presented in Figure SM2 .

我们修改后的三维提升网络的示意图如图 SM2 所示。

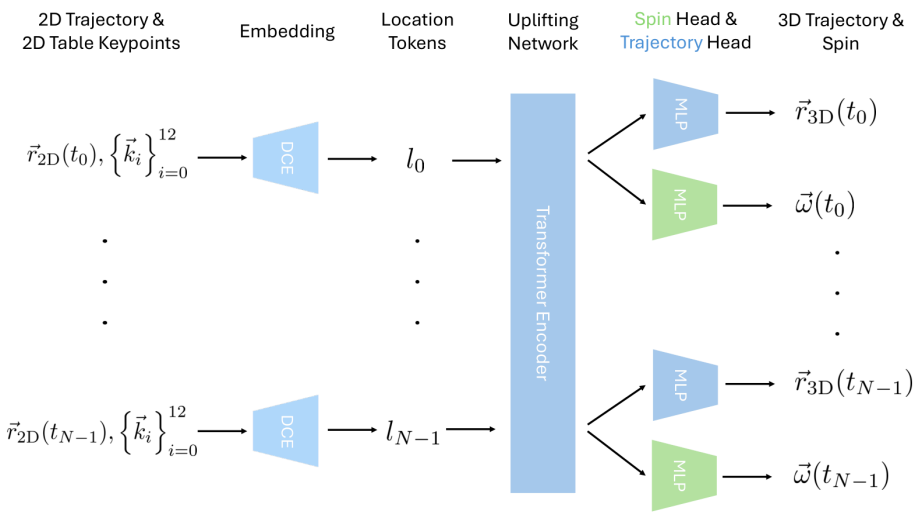


Figure SM2 .

图SM2。

Schematic illustration of the Lifting Network.

三维提升网络示意图。

First, the detected ball positions $\vec{r}_{2D}(t_n)$ and table keypoints $\{\vec{k}_i\}_{i=0}^{12}$ are embedded to obtain the sequence of location tokens $\{l_n\}_{n=0}^{N-1}$.

首先，嵌入检测到的球位置$\vec{r}_{2D}(t_n)$和球台关键点$\{\vec{k}_i\}_{i=0}^{12}$以获得位置词元序列$\{l_n\}_{n=0}^{N-1}$。

The lift network transforms these tokens and each output is further processed by a Spin Head to obtain the spin vector $\vec{\omega}(t_n)$ and Trajectory Head to obtain the 3D position $\vec{r}_{3D}(t_n)$ for each timestep t_n .

提升网络转换这些标记，每个输出由旋转头进一步处理以获得旋转向量$\vec{\omega}(t_n)$和轨迹头以获得每个时间步t_n的3D位置$\vec{r}_{3D}(t_n)$。

We refer to (Kienzle et al., 2026) for further information on the transformer architecture.

有关 Transformer 架构的更多信息，请参阅（Kienzle et al., 2026）。

A high-level architectural diagram of the Full-Sequence Lifting Network.

全序列三维提升网络的高级架构图。

The process starts on the left with inputs of 2D trajectories and table keypoints.

该过程从左侧开始，输入 2D 轨迹和球台关键点。

These are passed through a Disentangled Context Embedding (DCE) block to produce a sequence of location tokens (l_0 to l_{N-1}).

它们通过解耦上下文嵌入 (DCE) 块传递，以生成位置词元序列（l_0 到 l_{N-1}）。

These tokens enter a central "Uplifting Network" block, identified as a Transformer Encoder.

这些词元输入中央的三维提升网络，其核心为 Transformer 编码器。

The output tokens from this encoder are then split into parallel heads: a Trajectory Head and a Spin Head, both represented as MLP blocks.

然后，该编码器的输出标记被分成并行头：轨迹头和旋转头，两者都表示为 MLP 块。

The final outputs on the right are the 3D ball positions $r_{3D}(t_n)$ and 3D spin vectors $\omega(t_n)$ for each corresponding timestep.

右侧的最终输出是每个相应时间步长的 3D 球位置 $r_{3D}(t_n)$ 和 3D 旋转向量 $\omega(t_n)$。

B.2. Synthetic Dataset Generation

B.2.合成数据集生成

A network designed to process full points requires training data that reflects this new, more complex structure.

设计用于处理完整回合的网络需要反映这种新的、更复杂的结构的训练数据。

In contrast, previous methods (Kienzle et al., 2025 ; Kienzle et al., 2026) are only trained on smaller datasets of only 50k-140k individual segments, which can only be obtained from unreliable 2D-based time segmentation during inference.

相比之下，既有方法（Kienzle 等，2025、2026）仅用 5 万至 14 万个独立击球片段训练；推理时，这些片段还须由不可靠的二维时间分段得到。

To power our lifting network, we generate a new massive-scale synthetic dataset of approximately 3 million points , with each point consisting of multiple stitched segments.

为训练三维提升网络，我们生成约 300 万个合成回合，每个回合由多个轨迹片段拼接而成。

This dataset utilizes the MuJoCo (Todorov et al., 2012) physics simulation environment, ensuring all trajectories are physically correct and include perfect, per-frame ground truth for 3D position, and spin.

该数据集利用 MuJoCo（Todorov 等人，2012）物理模拟环境，确保所有轨迹在物理上都是正确的，并包括 3D 位置和旋转的完美每帧真值。

Simulating 3 million complete end-to-end points from purely random initial conditions is computationally infeasible.

直接从完全随机的初始条件仿真 300 万个完整回合，在计算上不可行。

To overcome this, we present a "stitching" pipeline, detailed in Algorithm 2 , that assembles points from a large pool of pre-validated, physically-correct segments.

为此，我们提出算法 2 所示的拼接流程，从已验证且符合物理规律的大片段池中组装完整回合。

The authors of (D’ Ambrosio et al., 2025) and (Kienzle et al., 2026) provide a large collection of initial conditions for serves and standard shots, which we utilize as a starting point for our pool.

(D’ Ambrosio et al., 2025) 和 (Kienzle et al., 2026) 的作者提供了大量发球和标准击球的初始条件，我们将其作为我们池的起点。

Simulating a full point then becomes a recursive, hybrid process: First, we simulate an initial segment (e.g., a serve) and find its final 3D position $\vec{r}_{N-1}^{3D}$.

完整回合的仿真成为递归的混合过程：首先模拟初始片段（例如发球），得到其末端三维位置 $\vec{r}_{N-1}^{3D}$ 。

This becomes the initial position for the next segment.

这将成为下一段的初始位置。

To determine the segments initial spin and velocity, we find the segment in our data pool whose initial position is the closest match to our current state.

为了确定段的初始旋转和速度，我们在数据池中找到初始位置与当前状态最接近的段。

We then simulate the trajectory and check if it is a valid table tennis shot segment (i.e., it clears the net and bounces on the opponent’ s side).

然后我们模拟轨迹并检查它是否是有效的乒乓球击球片段（即，它过网并在对手一侧弹跳）。

If valid, we "stitch" this new segment to our point and repeat the process until the point is complete.

若新片段有效，就将其拼接至当前回合，并重复这一过程，直至回合完成。

Furthermore, we identify that a major failure point for generalization to real-world data is the pre-serve ball toss , a phase that is difficult to clip before lifting.

此外，我们发现推广到现实世界数据的一个主要失败点是发球前抛球，这是一个在三维提升之前很难剪辑的阶段。

To make our network robust to this, we explicitly simulate an entirely new pool of synthetic "throw" segments.

为了使我们的网络对此具有鲁棒性，我们明确模拟了一个全新的合成“投掷”段池。

Our final training points are assembled by first simulating a "throw" segment, stitching a "serve" segment from our pool to its apex, and then recursively stitching the subsequent return segments.

最终的训练回合先模拟抛球，在最高点拼接发球片段，再递归拼接后续各次回球片段。

This novel, training data is the key that enables our network to learn the complex transitions between segments and handle unsegmented real-world videos.

这种新颖的训练数据是使我们的网络能够学习分段之间的复杂转换并处理未分段的现实世界视频的关键。

B.3. Ball Occlusion Model

B.3.球遮挡模型

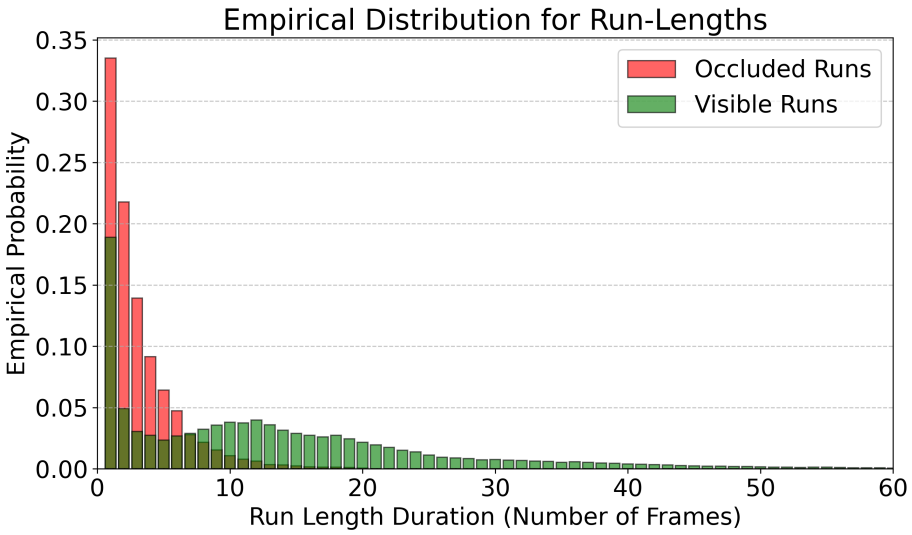


Figure SM3 .

图SM3。

Empirical distribution of contiguous occluded/non-occluded segment lengths.

连续遮挡片段与连续可见片段长度的经验分布。

During training, we randomly mask valid 2D detections using an empirically-derived occlusion model instead of a naive occlusion pattern (e.g. uniform random drop).

在训练过程中，我们使用经验得出的遮挡模型而不是朴素的遮挡模式（例如均匀随机下降）随机屏蔽有效的 2D 检测。

We record the frequency of consecutive visible and occluded frames seen in the broadcast videos (illustrated in Figure SM3).

我们记录转播视频中连续可见帧和遮挡帧的频率（如图 SM3 所示）。

To generate a random mask, we alternate between the two distributions and sample sequences of visible and occluded segment lengths.

生成随机掩膜时，交替从可见长度分布和遮挡长度分布中采样，组成可见与遮挡片段序列。

Appendix C 3D Hit-Point Identification Heuristics

附录 C 3D 击球点识别启发法

As described in the main paper, we identify hit-points and bounces by finding extrema in the 3D trajectory.

正如主论文中所述，我们通过查找 3D 轨迹中的极值来识别击中点和弹跳。

This 1D signal analysis is simple and robust.

这种一维信号分析简单而稳健。

To avoid identifying incorrect local extrema (e.g., from minor network noise) as valid events, we apply simple heuristics.

为了避免将不正确的局部极值（例如，来自较小的网络噪声）识别为有效事件，我们应用简单的启发式方法。

A 3D x-trajectory extremum is only considered a valid hit point if:

仅在以下情况下，3D x 轨迹极值才被视为有效击球点：

- The time between it and the previous hit point of the same type (e.g., peak-to-peak) is at least 0.2 seconds.

  它与相同类型的前一个击球点之间的时间（例如峰到峰）至少为 0.2 秒。

- The absolute value of its x-coordinate (distance from the center net) is at least 0.3 meters, ensuring it is a full shot segment and not a small jitter.

  其x坐标（距中心网的距离）的绝对值至少为0.3米，保证是完整的击球段，且没有小幅度的抖动。

A similar heuristic is applied to the z-trajectory for bounce detection.

类似的启发式方法应用于 z 轨迹以进行弹跳检测。

These simple filters are highly effective at isolating the true game events.

这些简单的过滤器对于隔离真实的游戏事件非常有效。

Appendix D Table Tennis Physics

附录 D 乒乓球物理

The physics of table tennis has been accurately modeled and can be used to predict or simulate ball trajectories.

乒乓球的物理原理已被精确建模，可用于预测或模拟球轨迹。

In this section, we go over the different physical models used in the paper.

在本节中，我们将回顾本文中使用的不同物理模型。

D.1. Aerodynamics

D.1.空气动力学

The ball’ s trajectory is defined by the following ODE :

球的轨迹由以下 ODE 定义：

(SM3)
$$m\boldsymbol{a} = \underbrace{-k_d\|\boldsymbol{v}\|\boldsymbol{v}}_{\text{Drag}} + \underbrace{k_m\boldsymbol{\omega}\times\boldsymbol{v}}_{\text{Magnus force}} + \boldsymbol{g}$$

where $k_d = 3.8 \times 10^{-4}$ is the drag coefficient and $k_m = 3 \times 10^{-6}$ is the Magnus coefficient and $\boldsymbol{g}$ is the gravity vector.

其中$k_d = 3.8 \times 10^{-4}$为阻力系数，$k_m = 3 \times 10^{-6}$为马格努斯系数，$\boldsymbol{g}$为重力矢量。

D.2. Table bounce

D.2。球台反弹

The table bounce is defined by the following equation:

球与台面反弹的关系由下式定义：

(SM4)
$$\begin{aligned}\boldsymbol{v}^+ &= \boldsymbol{A}\boldsymbol{v}^- + \boldsymbol{B}\boldsymbol{\omega}^-\\ \boldsymbol{\omega}^+ &= \boldsymbol{C}\boldsymbol{v}^- + \boldsymbol{D}\boldsymbol{\omega}^-\end{aligned}$$

The matrices $\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C}, \boldsymbol{D}$ encode restitution and friction parameters.

矩阵$\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C}, \boldsymbol{D}$编码恢复和摩擦参数。

It distinguishes two cases: the ball can either have a rolling contact or a sliding contact.

它区分两种情况：球可以具有滚动接触或滑动接触。

The nature of the contact is determined by the coefficient:

接触的性质由系数决定：

(SM5)
$$\begin{aligned} \alpha &= \frac{\mu(1+\mathrm{COR})\,|v_z^-|}{v_s}, \\ v_s &= \sqrt{\left(v_x^- + \omega_y^- r\right)^2 + \left(v_y^- + \omega_x^- r\right)^2} \end{aligned}$$

where COR is the coefficient of restitution, μ is the friction coefficient between the ball and the table and v_s is the tangential velocity of the ball at the point of contact with the table surface.

其中COR为恢复系数，μ为球与台面之间的摩擦系数，v_s为球与台面接触点的切向速度。

If $\alpha \geqslant 0.4$, then the velocity of the ball’ s contact point is 0 and the ball is rolling:

若$\alpha \geqslant 0.4$，则球接触点的速度为0，球在滚动：

(SM6)
$$\begin{aligned} \boldsymbol{A} &= \begin{bmatrix} 1-\alpha & 0 & 0 \\ 0 & 1-\alpha & 0 \\ 0 & 0 & -COR \end{bmatrix} \quad \boldsymbol{B} = \begin{bmatrix} 0 & \alpha r & 0 \\ -\alpha r & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \\ \boldsymbol{C} &= \begin{bmatrix} 0 & -\frac{3\alpha}{2r} & 0 \\ \frac{3\alpha}{2r} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \boldsymbol{D} = \begin{bmatrix} 1-\frac{3}{2}\alpha & 0 & 0 \\ 0 & 1-\frac{3}{2}\alpha & 0 \\ 0 & 0 & 1 \end{bmatrix} \end{aligned}$$

If $\alpha < 0.4$, then the velocity of the ball’ s contact point is not 0 and the ball is sliding:

若$\alpha < 0.4$ ，则小球接触点的速度不为0，小球正在滑动：

(SM7)
$$\begin{aligned} \boldsymbol{A} &= \begin{bmatrix} 0.6 & 0 & 0 \\ 0 & 0.6 & 0 \\ 0 & 0 & -COR \end{bmatrix} \quad \boldsymbol{B} = \begin{bmatrix} 0 & 0.4r & 0 \\ -0.4r & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \\ \boldsymbol{C} &= \begin{bmatrix} 0 & -0.6\,/\,r & 0 \\ 0.6\,/\,r & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \boldsymbol{D} = \begin{bmatrix} 0.4 & 0 & 0 \\ 0 & 0.4 & 0 \\ 0 & 0 & 1 \end{bmatrix} \end{aligned}$$

Appendix E ODE Fit

附录 E ODE 拟合

Given the reconstructed 3D ball trajectory $\{\vec{r}_{3D}(t_n)\}_{n=0}^{N-1}$ produced by our lifting network, we recover a physically consistent trajectory by estimating the initial ball state that best explains the observations under the physics model from Appendix D .

给定我们的三维提升网络生成的重建的 3D 球轨迹$\{\vec{r}_{3D}(t_n)\}_{n=0}^{N-1}$，我们通过估计最能解释附录 D 物理模型下的观察结果的初始球状态来恢复物理一致的轨迹。

We simulate ball flight using the aerodynamic ODE from Equation SM3 together with the table–bounce model in Equations SM4 and SM5 , integrating with a fixed-step RK4 (RK4) scheme.

仿真使用式 SM3 的空气动力学常微分方程，以及式 SM4、SM5 的触台反弹模型，通过固定步长四阶龙格—库塔法（RK4）积分。

Table impacts are handled explicitly: at each step, we detect when the height crosses the table plane $z = h_{\text{table}} = 0.78\,m$ with downward velocity, interpolate to the exact impact time, snap the ball to the surface, and apply the corresponding sliding or rolling update from Equations SM6 and SM7 .

显式处理触台碰撞：逐步检测球是否向下穿过台面 $z = h_{\text{table}} = 0.78\,m$，通过插值求出准确碰撞时刻，将球位置置于台面，再按式 SM6、SM7 更新滑动或滚动接触状态。

The simulation is limited to at most two bounces, which covers all trajectories encountered in practice.

模拟仅限于最多两次反弹，涵盖了实践中遇到的所有轨迹。

Let $\vec{x}_0 = [\vec{p}_0^{\top}, \vec{v}_0^{\top}, \vec{\omega}_0^{\top}]^{\top}$ denote the unknown initial position, velocity, and spin at time t_0 .

令$\vec{x}_0 = [\vec{p}_0^{\top}, \vec{v}_0^{\top}, \vec{\omega}_0^{\top}]^{\top}$表示$t_0$时刻的未知初始位置、速度和旋转。

For any candidate $\vec{x}_0$, the physics model generates a simulated 3D trajectory

对于任何候选$\vec{x}_0$，物理模型都会生成模拟的 3D 轨迹

$$\hat{\vec{r}}_{3D}(t_n, \vec{x}_0),$$

which we align to the network predictions $\vec{r}_{3D}(t_n)$ through a robust nonlinear least-squares objective:

我们通过稳健的非线性最小二乘目标将其与网络预测$\vec{r}_{3D}(t_n)$对齐：

$$
\begin{aligned}
\min_{\vec{p}_0,\,\vec{v}_0,\,\vec{\omega}_0} \quad & \sum_{n\,\in\,\mathcal{I}} \rho\Big(\big\| \hat{\vec{r}}_{3\mathrm{D}}(t_n, \vec{p}_0, \vec{v}_0, \vec{\omega}_0) - \vec{r}_{3\mathrm{D}}(t_n) \big\|_2 \Big) \\
\text{s.t.} \quad & \vec{p}_{\min} \;\le\; \vec{p}_0 \le\; \vec{p}_{\max}, \\
& \|\vec{v}_0\|_\infty \;\le\; v_{\max}, \\
& \|\vec{\omega}_0\|_\infty \;\le\; \omega_{\max}.
\end{aligned}
\tag{SM8}
$$

Here, $\mathcal{I}$ is the set of valid (non-missing) observations and $\rho(\cdot)$ is a Huber loss that downweights outliers.

这里，$\mathcal{I}$是有效（非缺失）观测值的集合，$\rho(\cdot)$是降低异常值权重的 Huber 损失。

The bounds ensure that the recovered initial state remains physically plausible.

这些界限确保恢复的初始状态在物理上保持合理。

Solving this problem yields the optimal state

解决这个问题会产生最优状态

$$
\vec{x}_0^{\,\star} = [\vec{p}_0^{\,\star}, \vec{v}_0^{\,\star}, \vec{\omega}_0^{\,\star}],
$$

together with a fully simulated, physically consistent trajectory $\hat{\vec{r}}_{3\mathrm{D}}(t_n, \vec{x}_0^{\,\star})$.

以及完全模拟的、物理上一致的轨迹$\hat{\vec{r}}_{3\mathrm{D}}(t_n, \vec{x}_0^{\,\star})$。

For each segment, we report the root-mean-square error between simulated and reconstructed positions, as well as the number of predicted bounces, which allows us to distinguish single- and multi-bounce trajectories in downstream analyses.

对于每个片段，我们报告仿真位置与重建位置之间的均方根误差，以及预测的弹跳次数，以便在后续分析中区分单次弹跳与多次弹跳轨迹。

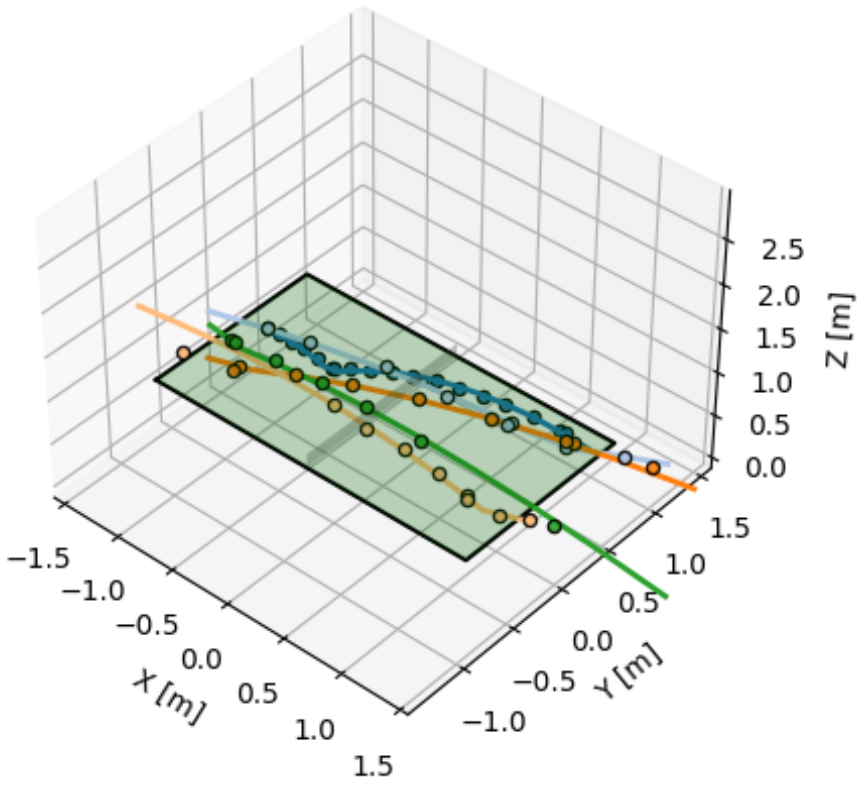


Figure SM4 .

图SM4。

Fitting the ODE constrained ball trajectory to the 3D points inferred by the Lifting Network.

将受 ODE 约束的物理轨迹拟合至三维提升网络输出的三维点。

A 3D visualization showing multiple ball trajectory segments over a green table tennis table.

3D 可视化显示绿色乒乓球桌上的多个球轨迹段。

The plot displays discrete circular markers representing the 3D points inferred by the Lifting Network, which are closely fitted by continuous, color-coded solid lines representing the physics-based ODE constrained trajectories.

图中圆点表示三维提升网络预测的位置，不同颜色的连续实线表示拟合得到的 ODE 物理轨迹。

The trajectories illustrate different types of shots, including those that clear the net and bounce on the table surface.

这些轨迹展示了不同类型的击球，包括过网和在桌面上弹跳的击球。

Appendix F Racket Strike Reconstruction

附录 F 球拍击球重建

Given the pre- and post-impact ball velocity and spin, the impulse delivered by the racket is fully determined.

给定击球前后球的线速度与旋转状态，即可确定球拍施加的冲量。

This allows us to recover the racket’ s orientation $\mathbf{R}_{\mathrm{r}}^{\mathrm{w}}$ and velocity $\mathbf{V}_{\mathrm{r}}^{\mathrm{w}}$ at contact.

这使我们能够恢复球拍接触时的方向$\mathbf{R}_{\mathrm{r}}^{\mathrm{w}}$和速度$\mathbf{V}_{\mathrm{r}}^{\mathrm{w}}$。

Although multiple solutions may exist in principle (Liu et al., 2012) , any recovered pair $(\mathbf{R}_{\mathrm{r}}^{\mathrm{w}}, \mathbf{V}_{\mathrm{r}}^{\mathrm{w}})$ exactly reproduces the observed ball trajectory and is therefore consistent with the recorded impact.

尽管原则上可能存在多种解决方案（Liu et al., 2012），但任何恢复的一对$(\mathbf{R}_{\mathrm{r}}^{\mathrm{w}}, \mathbf{V}_{\mathrm{r}}^{\mathrm{w}})$都准确地再现了观察到的球轨迹，因此与记录的撞击一致。

To compute $(\mathbf{R}_{\mathrm{r}}^{\mathrm{w}}, \mathbf{V}_{\mathrm{r}}^{\mathrm{w}})$, we formulate an OCP (OCP) that minimizes the L2 distance between the predicted and observed bounce locations.

为了计算$(\mathbf{R}_{\mathrm{r}}^{\mathrm{w}}, \mathbf{V}_{\mathrm{r}}^{\mathrm{w}})$，我们制定了一个 OCP (OCP)，它可以最小化预测和观察到的反弹位置之间的 L2 距离。

We use a single-shooting formulation with an $\mathrm{RK4}$ integrator, enabling us to propagate the full ball-flight ODE, including the Magnus effect, unlike the simplified models used in (Liu et al., 2012) .

我们采用单次打靶法（single shooting），结合积分器 $\mathrm{RK4}$，求解包含马格努斯效应的完整飞行动力学方程；这不同于 Liu 等人（2012）的简化模型。

Racket–ball contact is modeled using the linear impact formulation of (Nakashima et al., 2010) , expressed in the racket reference frame,

球拍与球的接触使用线性冲击公式（Nakashima et al., 2010）进行建模，以球拍参考系表示，

(SM9)
$$\begin{aligned} \boldsymbol{v}_r^+ &= \boldsymbol{A}\,\boldsymbol{v}_r^- + \boldsymbol{B}\,\boldsymbol{\omega}_r^-, \\ \boldsymbol{\omega}_r^+ &= \boldsymbol{C}\,\boldsymbol{v}_r^- + \boldsymbol{D}\,\boldsymbol{\omega}_r^-, \end{aligned}$$

where $\boldsymbol{v}_r^-$ and $\boldsymbol{\omega}_r^-$ denote the incoming linear and angular ball velocities in the racket frame, and $\boldsymbol{v}_r^+$, $\boldsymbol{\omega}_r^+$ are the outgoing quantities.

其中 $\boldsymbol{v}_r^-$ 和 $\boldsymbol{\omega}_r^-$ 分别为球拍坐标系中的入射线速度与角速度，$\boldsymbol{v}_r^+$ 和 $\boldsymbol{\omega}_r^+$ 为出射时的对应量。

The model was originally developed for inverted-rubber rackets, which dominate competitive play; we therefore use a single fixed parameter set in all experiments.

该模型最初针对竞技比赛常用的反胶球拍开发，因此所有实验均采用同一组固定参数。

The matrices used in the model are defined as follows:

模型中使用的矩阵定义如下：

(SM10)

$$
\begin{aligned}
\boldsymbol{A} &= \begin{bmatrix} 1-\frac{k_p}{m} & 0 & 0 \\ 0 & 1-\frac{k_p}{m} & 0 \\ 0 & 0 & -COR \end{bmatrix}, \quad \boldsymbol{B} = \frac{k_p}{m}\begin{bmatrix} 0 & r & 0 \\ -r & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \\
\boldsymbol{C} &= \frac{k_p}{I}\begin{bmatrix} 0 & -r & 0 \\ r & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad \boldsymbol{D} = \begin{bmatrix} 1-\frac{k_p}{I}r^2 & 0 & 0 \\ 0 & 1-\frac{k_p}{I}r^2 & 0 \\ 0 & 0 & 1 \end{bmatrix}
\end{aligned}
$$

where $I = 2\ /\ 3r^2m$ is the inertia of a hollow sphere, $m = 2.7g$ is the ball mass, $r = 0.02m$ is the ball radius, $COR = 0.75$ is the Coefficient Of Restitution of the racket, and $k_p = 0.002$ is the friction coefficient of the racket.

其中 $I = 2\ /\ 3r^2m$ 为空心球的转动惯量，$m = 2.7g$ 为球质量，$r = 0.02m$ 为半径，$COR = 0.75$ 为球拍恢复系数，$k_p = 0.002$ 为球拍摩擦系数。

The racket state is represented by its position $\boldsymbol{p}_r$, world-frame orientation R_r^w, and linear velocity $\boldsymbol{V}_r^w$.

球拍状态由其位置$\boldsymbol{p}_r$、世界坐标系方向R_r^w和线速度$\boldsymbol{V}_r^w$来表示。

Transforming ball velocities from the world frame to the racket frame yields

将球速度从世界坐标系转换为球拍坐标系产生

(SM11)

$$
\begin{cases}
\boldsymbol{v}_r &= \left(R_r^w\right)^\top \left(\boldsymbol{v}_b^w - \boldsymbol{V}_r^w\right), \\
\boldsymbol{\omega}_r &= \left(R_r^w\right)^\top \boldsymbol{\omega}_b^w,
\end{cases}
$$

In contrast to (Liu et al., 2012) , who estimate racket parameters using a simplified flight model that neglects the Magnus effect and the vertical drag component, we solve the boundary-value problem using the full aerodynamic flight dynamics.

与（Liu et al., 2012）使用忽略马格努斯效应和垂直阻力分量的简化飞行模型估计球拍参数相比，我们使用完整的空气动力飞行动力学来解决边值问题。

We simulate the outgoing trajectory with a RK4 integrator, combining the racket–ball interaction model with the aerodynamic ODEs.

我们使用 RK4 积分器模拟传出轨迹，将球拍-球相互作用模型与空气动力学常微分方程相结合。

We formulate the reconstruction of racket parameters as a nonlinear OCP with single shooting:

采用单次打靶法，将球拍参数重建表述为非线性最优控制问题（OCP）：

$$
\min_{t_{\text{net}},\, \boldsymbol{q}_r,\, \boldsymbol{V}_r^w} \quad \alpha \left\| \boldsymbol{\omega}_b^+ - \boldsymbol{\omega}_{\text{tgt}} \right\|^2 + \beta \left\| \boldsymbol{p}_b(t_{\text{flight}}) - \boldsymbol{p}_{\text{tgt}} \right\|^2
$$

(SM12) $$\text{s.t.}\quad \boldsymbol{e}_z^\top \boldsymbol{p}_b(t_{\text{flight}}) = 0,$$

(SM13) $$\|\boldsymbol{q}_r\|^2 = 1,$$

$$\big(R(\boldsymbol{q}_r)\boldsymbol{e}_z\big)^\top \boldsymbol{V}_r^w \geq 0,$$

(SM14) $$\boldsymbol{e}_y^\top R(\boldsymbol{q}_r)\boldsymbol{e}_z \geq 0.$$

Here, t_{net} is the net-crossing time, t_{flight} the bounce time on the opponent’ s side, $\boldsymbol{q}_r$ the racket orientation (unit quaternion), $\boldsymbol{\omega}_b^+$ the outgoing spin after impact, $\boldsymbol{\omega}_{\text{tgt}}$ the target spin, and $\boldsymbol{p}_{\text{tgt}}$ the desired bounce location.

其中，t_{net}为过网时间，t_{flight}为对方弹跳时间，$\boldsymbol{q}_r$球拍方向（单位四元数），$\boldsymbol{\omega}_b^+$击球后的出旋，$\boldsymbol{\omega}_{\text{tgt}}$目标旋转，$\boldsymbol{p}_{\text{tgt}}$所需的弹跳位置。

The weights α and β balance spin accuracy and landing precision, and ε enforces a safety clearance above the net.

权重 α 和 β 用于权衡旋转与落点精度，ε 用于约束球越网时的安全高度。

The resulting nonlinear OCP is solved using CasADi (Andersson et al., 2019) together with IPOPT (Wächter and Biegler, 2006) , using a single-shooting transcription of the dynamics.

通过单次打靶法离散化动力学，使用 CasADi（Andersson 等，2019）和 IPOPT（Wächter 和 Biegler，2006）求解该非线性最优控制问题。

We run it with a fixed number of nodes and enforce that the last node be the bouncing position with Equation SM12 .

我们使用固定数量的节点运行它，并使用方程 SM12 强制最后一个节点为弹跳位置。

We also constrain the racket to always be facing the table with Equation SM14 and the racket’ s norm to always be in the same direction as the ball’ s outgoing velocity Equation SM13 .

式 SM14 约束球拍始终朝向球台，式 SM13 约束拍面法向与球的出射速度朝向一致。

We validate our solver with a Monte-Carlo evaluation in which we randomly sample hit positions, bounce locations, incoming velocities, spins, and flight times over realistic ranges.

我们通过蒙特卡罗评估来验证我们的求解器，其中我们在实际范围内随机采样命中位置、弹跳位置、传入速度、旋转和飞行时间。

Across $10{,}000$ trials, the method converges to an exact solution (sub-millimeter bounce error) in 97.22% of cases.

在$10{,}000$试验中，该方法在 97.22% 的情况下收敛到精确解（亚毫米弹跳误差）。

Appendix G Filtering Details

附录 G 筛选细节

Table SM1 .

表 SM1。

Quality criteria used across reconstruction and filtering stages.

重建和过滤阶段使用的质量标准。

Stage 阶段	Criterion / Threshold 标准/阈值
2D Reprojection Check 2D重投影检查	
– Max normalized error – 最大归一化误差	0.2
3D-Domain Filtering 3D 域过滤	
– Max ODE fit RMSE – 最大 ODE 拟合 RMSE	0.3 m
– Human average transl bounds: – 人体平均平移位置范围：	
$\lvert\bar{h}_x\rvert$	$\in [0.5,\ 8.22]$ m
$\lvert\bar{h}_y\rvert$	$\in [0.5,\ 1.628]$ m
Hit-Point Identification 击中点识别	
– Min peak-to-peak time – 最短峰峰时间	0.2 s
– Min extrema x position – 最小极值x -位置	0.3 m
– Min number of hits – 最少击球次数	2

We discuss the filtering of our dataset in Section 3.6 of the main paper.

我们在主论文的 3.6 节中讨论了数据集的过滤。

We provide the individual threshold used in each filtering step in Table SM1 .

我们在表 SM1 中提供了每个过滤步骤中使用的单独阈值。

G.1. Filtering Bias

G.1。筛选偏差

There is no obvious pattern to the type of rejected clips at the “Gameplay clip" stage.

在“游戏剪辑”阶段被拒绝的剪辑类型没有明显的模式。

Among the filtered clips, 204,720 are singles, 86,247 are doubles clips, and the distribution of the camera view is no different from the curated dataset (see Figures SM6 , SM7 , SM8).

在过滤后的剪辑中，204,720是单打剪辑，86,247是双打剪辑，并且摄像机视图的分布与策划的数据集没有什么不同（见图SM6，SM7，SM8）。

One common reason for clip rejection stems from the previous “Scoreboard clips" stage: the scoreboard updates are not always a reliable indicator of starts and ends of gameplay (i.e. the scoreboard may be updated during the next point).

一种常见的剔除原因来自上一阶段：记分牌更新并不总与回合起止同步，有时会在下一回合进行中才更新。

Our heuristic in the “Gameplay clip" stage is rather conservative, focusing on reducing false positives.

我们在“游戏剪辑”阶段的启发式相当保守，专注于减少误报。

Importantly, the final dataset is not skewed toward clean trajectories.

重要的是，最终的数据集并不偏向干净的轨迹。

In Figure 8, the inter-hit ball times indicate highly varied gameplay speeds, and the dataset also includes clips with high occlusion rates (Figure SM5).

在图 8 中，击球间时间表明游戏速度差异很大，并且数据集还包括具有高遮挡率的剪辑（图 SM5）。

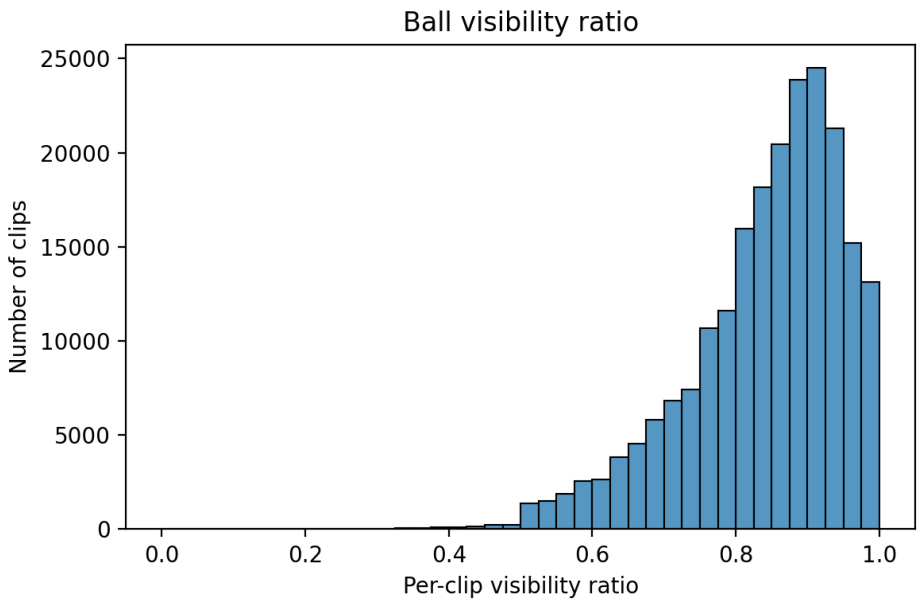


Figure SM5 .

图SM5。

Distribution of the fraction of frames where the ball is visible.

各片段中球可见帧占比的分布。

Appendix H Additional Dataset Statistics

附录 H 其他数据集统计数据

We provide further insights into the TT4D dataset by analyzing the distribution of the estimated camera parameters and the spatial statistics of the trajectories generated by our Flow Matching model.

我们通过分析估计的相机参数的分布以及我们的流匹配模型生成的轨迹的空间统计数据，提供对 TT4D 数据集的进一步见解。

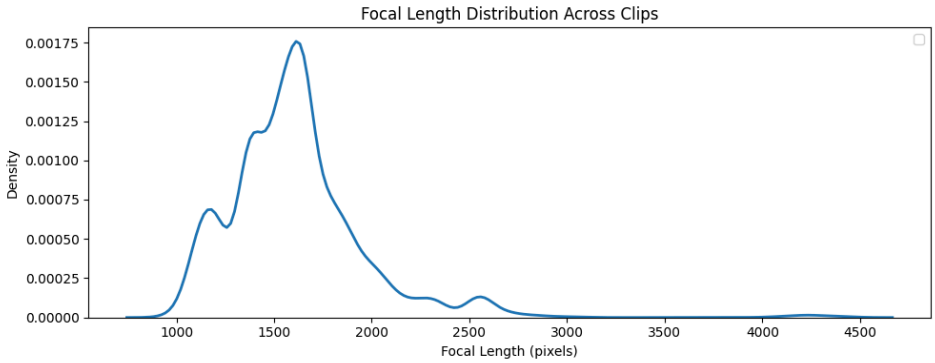


Figure SM6 .

图SM6。

Distribution of camera focal lengths in the TT4D dataset.

TT4D 数据集中相机焦距的分布。

A density plot showing the distribution of camera focal lengths.

显示相机焦距分布的密度图。

The horizontal axis represents focal length in millimeters, ranging from under 1000 to over 4500.

横轴代表焦距（以毫米为单位），范围从 1000 以下到 4500 以上。

The vertical axis represents density.

纵轴表示密度。

The curve is right-skewed with a primary peak around 1600 mm, a secondary smaller peak near 1200 mm, and a long tail extending to the right.

该曲线右偏，主峰位于 1600 毫米附近，次要峰位于 1200 毫米附近，并且有一条向右延伸的长尾。

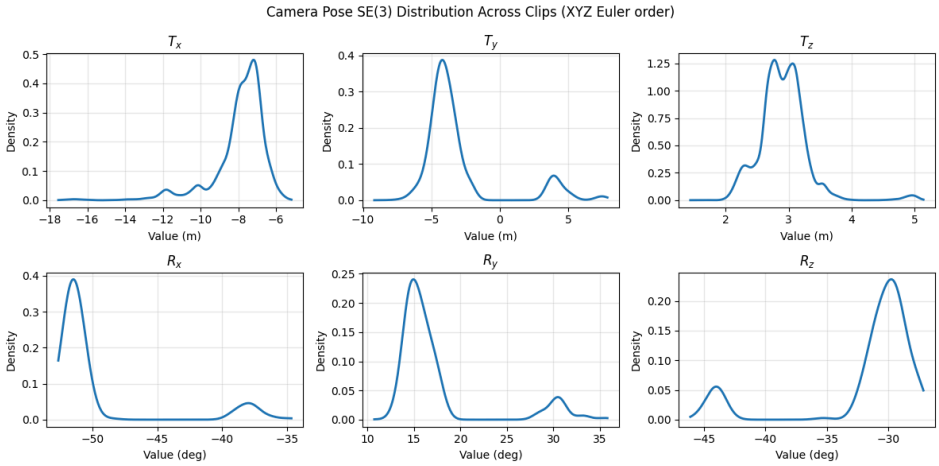


Figure SM7 .

图SM7。

The distribution of camera poses $SE(3)$ in the TT4D dataset reveals two most common camera configurations, where one is roughly 5x more common than the other.

TT4D 数据集中相机位姿的分布$SE(3)$揭示了两种最常见的相机配置，其中一种配置比另一种配置常见约 5 倍。

A 2 by 3 grid of density plots illustrating the distribution of camera poses.

2 x 3 网格的密度图说明了相机位姿的分布。

The top row plots translation parameters (T_x, T_y, T_z) in meters, while the bottom row plots rotation parameters (R_x, R_y, R_z) in degrees.

顶行以米为单位绘制平移参数（T_x、T_y、T_z），而底行以度为单位绘制旋转参数（R_x、R_y、R_z）。

Most of the individual subplots display a clear bimodal distribution featuring one prominent major peak and one significantly smaller secondary peak.

大多数单独的子图显示出清晰的双峰分布，具有一个突出的主峰和一个明显较小的次峰。





Figure SM8 .

图SM8。

Two common camera poses in the TT4D dataset.

TT4D 数据集中两个常见的相机位姿。

The top and bottom figures corresponds to Euler angle modes of $(-55°, 15°, -30°)$ and $(-37°, 32°, -42°)$, respectively.

上下两图分别对应欧拉角分布中位于 $(-55°, 15°, -30°)$ 和 $(-37°, 32°, -42°)$ 的众数。

Two photographs of table tennis matches demonstrating different camera angles.

两张展示不同拍摄角度的乒乓球比赛照片。

The top image shows a doubles match viewed from a higher, downward-angled perspective, corresponding to Euler angles of -55, 15, and -30 degrees.

上图显示了从更高的向下角度观看的双打比赛，对应于 -55、15 和 -30 度的欧拉角。

Four players are visible around a blue table on a red floor.

红色地板上的蓝色球台周围可见四名球员。

The bottom image shows a singles match from a slightly lower, more offset perspective, corresponding to Euler angles of -37, 32, and -42 degrees.

下图显示了从稍低、更偏移的角度观看的单打比赛，对应于 -37、32 和 -42 度的欧拉角。

Two players and a seated umpire are visible around a blue table on a gray floor.

灰色地板上的蓝色球台周围可以看到两名球员和一名就座的裁判。

Figure SM6 shows a diverse distribution of different focal lengths in our dataset.

图 SM6 显示了我们的数据集中不同焦距的多样化分布。

The distribution of the extrinsic camera parameters (translation $\mathbf{T}$ and rotation $\boldsymbol{\theta}$ in Euler angles) is visualized in Figure SM7 .

图 SM7 展示相机外参的分布，包括平移 $\mathbf{T}$，以及以欧拉角表示的旋转 $\boldsymbol{\theta}$。

While a wide variety of poses is present in the dataset, a bimodal distribution is visible in some parameters, indicating that some camera poses are especially frequent in the dataset.

虽然数据集中存在各种各样的姿态，但在某些参数中可以看到双峰分布，这表明某些相机位姿在数据集中特别频繁。

We show two dominant views in Figure SM8

图 SM8 展示了两种主要拍摄视角。

Appendix I Generative Model Details

附录 I 生成模型详细信息

I.1. Data and Representation

I.1.数据和表示

Let $\mathcal{D}$ denote our dataset of reconstructed trajectories.

让$\mathcal{D}$表示我们的重建轨迹数据集。

Each trajectory $\tau = (o_1, ..., o_N) = o_{1:N} \in \mathcal{D}$ consists of observations $o_t = (b_t, h_{1,t}, h_{2,t})$, where the ball state is $b_t \in \mathbb{R}^3$ and the skeletons $h_{1,t}, h_{2,t} \in \mathbb{R}^{21\times 3}$.

每条轨迹$\tau = (o_1, ..., o_N) = o_{1:N} \in \mathcal{D}$由观测值$o_t = (b_t, h_{1,t}, h_{2,t})$组成，其中球状态为$b_t \in \mathbb{R}^3$，骨骼状态为$h_{1,t}, h_{2,t} \in \mathbb{R}^{21\times 3}$。

All trajectories are resampled to 30 Hz.

所有轨迹均重新采样至 30 Hz。

In total, the dataset contains 237,054 reconstructed points (~ 151 hours).

该数据集共包含 237,054 个重建回合（ ~ 151 小时）。

I.2. Model Architecture and Training

I.2.模型架构和训练

We model the time-dependent vector field $v_\theta(\tau_t, t|c)$ using a DiT-style architecture with 6 attention heads, 6 layers, MLP ratio of 4, dropout of 0.1, model embedding size of 384,

conditioning embedding size of 512, and time embedding size of 128.

我们使用具有 6 个注意力头、6 层、MLP 比率为 4、dropout 为 0.1、模型嵌入大小为 384、条件嵌入大小为 512、时间嵌入大小为 128 的 DiT 式架构对时间相关向量场$v_\theta(\tau_t, t|c)$进行建模。

We predict a horizon of 20 observations $\tau = o_{1:20}$ from an observation history of $c = o_{-9:0}$ (Tong et al., 2024b ; Tong et al., 2024a) .

根据历史观测 $c = o_{-9:0}$，预测未来 20 个观测时刻的状态 $\tau = o_{1:20}$（Tong 等，2024a、2024b）。

We train this model for 600,000 iterations with a batch size of 512 on a single NVIDIA RTX 4090 GPU.

我们在单个 NVIDIA RTX 4090 GPU 上训练该模型进行 600,000 次迭代，批量大小为 512。

Optimization uses AdamW (Loshchilov and Hutter, 2019) with a learning rate of 2×10^{-4} , a cosine decay schedule, and 500 warmup steps.

优化使用 AdamW（Loshchilov 和 Hutter，2019），学习率为2×10^{-4}，余弦衰减时间表和 500 个预热步骤。

I.3. Generation Quality

I.3。生成质量

The model weights used at inference are computed by passing an exponential moving average (EMA) filter over the sequence of model weights from training .

推理中使用的模型权重是通过对训练中的模型权重序列传递指数移动平均 (EMA) 过滤器来计算的。

During evaluation, we generate future trajectories by numerically integrating the learned ODE forward from Gaussian noise using $n_{\text{sample}} = 5$ uniform steps.

评估时以高斯噪声为起点，用 $n_{\text{sample}} = 5$ 个等间隔步长对学习得到的 ODE 进行数值积分，生成未来轨迹。

We assess two categories of metrics: Physical Plausibility: We run the generated trajectories through the same Physics-Based ODE Fit filter from our pipeline’ s Stage 4 (Section 3.6).

我们评估两类指标： 物理合理性：我们通过流程第 4 阶段（第 3.6 节）中相同的基于物理的 ODE 拟合过滤器运行生成的轨迹。

As shown in Figure 8 in the main paper, the distribution of this error closely matches the real data.

正如主论文中的图 8 所示，该误差的分布与真实数据非常吻合。

We also measure the smoothness of ball accelerations, continuity of human joint velocities, and any violations of kinematic limits across the full predicted rallies.

我们还测量球加速度的平滑度、人体关节速度的连续性以及整个预测回合中任何违反运动学限制的情况。

Gameplay Realism and Diversity: We compute distributional statistics over sequence duration, inter-hit timing, ball height profiles, and stroke kinematics.

游戏真实性和多样性：我们计算序列持续时间、击球间时间、球高度轮廓和击球运动学的分布统计数据。

We compare these distributions from the generated rallies to those from a held-out test set of real data.

我们将生成的回合的这些分布与保留的真实数据测试集的分布进行比较。

Qualitative visualizations of sampled rallies (see Figure SM12) further demonstrate that the learned flow produces coherent long-horizon behavior, including consistent hitting mechanics and realistic ball-table interactions.

采样回合的定性可视化（见图 SM12）进一步表明，学习的流程会产生连贯的长视野行为，包括一致的击球机制和真实的球桌交互。

Appendix J Evaluation Metrics

附录 J 评估指标

We use several metrics to evaluate our Lifting Network , which are defined below.

我们使用几个指标来评估我们的三维提升网络，这些指标的定义如下。

Let M be the total number of trajectories in the test set, and N_m be the number of frames with valid 2D ball detections in the m -th trajectory.

令M为测试集中的轨迹总数，N_m为第m条轨迹中有效检测到二维球的帧数。

Let $\vec{r}(t_n)$ and $\vec{\omega}(t_n)$ be the ground truth 3D position and 3D spin at timestep t_n , and let $\hat{\vec{r}}(t_n)$ and $\hat{\vec{\omega}}(t_n)$ be the predicted values.

令$\vec{r}(t_n)$和$\vec{\omega}(t_n)$为t_n时间步的3D位置和3D旋转真值，并令$\hat{\vec{r}}(t_n)$和$\hat{\vec{\omega}}(t_n)$为预测值。

J.1. 3D Trajectory Error

J.1。 3D 轨迹误差

The $\Delta\vec{r}_{3D}$ metric computes the mean Euclidean distance between the predicted and ground truth 3D positions, measured in centimeters.

$\Delta\vec{r}_{3D}$度量计算预测位置和真实值 3D 位置之间的平均欧几里得距离（以厘米为单位）。

It is our primary metric for 3D accuracy on the synthetic and TT4DBench datasets.

它是我们在合成数据集和 TT4DBench 数据集上衡量 3D 准确性的主要指标。

(SM15)
$$\Delta\vec{r}_{3D} = \frac{1}{M}\sum_{m=0}^{M-1}\frac{1}{N_m}\sum_{n=0}^{N_m-1}\|\vec{r}_{3D}(t_n) - \hat{\vec{r}}_{3D}(t_n)\|_2$$

J.2. 3D Spin Error

J.2。 3D 旋转误差

The $\Delta\vec{\omega}$ metric computes the mean Euclidean distance between the predicted and ground truth 3D spin vectors, measured in Hz.

$\Delta\vec{\omega}$度量计算预测值和真实值 3D 旋转向量之间的平均欧几里得距离（以 Hz 为单位）。

It is only used on our synthetic dataset, as real-world datasets do not provide ground truth 3D spin vectors.

它仅用于我们的合成数据集，因为现实世界的数据集不提供真值 3D 旋转向量。

(SM16)
$$\Delta\vec{\omega} = \frac{1}{M}\sum_{m=0}^{M-1}\frac{1}{N_m}\sum_{n=0}^{N_m-1}\|\vec{\omega}(t_n) - \hat{\vec{\omega}}(t_n)\|_2$$

J.3. 2D Reprojection Error

J.3。 2D重投影误差

For real-world datasets like TTST that lack 3D ground truth, we evaluate the 2D reprojection error $\Delta\vec{r}_{2D}$.

对于像 TTST 这样缺乏 3D 真值的现实数据集，我们评估 2D 重投影误差$\Delta\vec{r}_{2D}$。

The predicted 3D trajectory $\hat{\vec{r}}(t_n)$ is projected back into the 2D image plane using the provided camera projection matrix $\mathcal{P}$, and compared against the 2D ground truth annotations $\vec{r}_{2D}(t_n)$.

使用提供的相机投影矩阵$\mathcal{P}$将预测的 3D 轨迹$\hat{\vec{r}}(t_n)$投影回 2D 图像平面，并与 2D 真值标注$\vec{r}_{2D}(t_n)$进行比较。

The error is measured in pixels.

误差以像素为单位进行测量。

(SM17) $$\Delta\vec{r}_{2D} = \frac{1}{M}\sum_{m=0}^{M-1}\frac{1}{N_m}\sum_{n=0}^{N_m-1}\|\mathcal{P}(\hat{\vec{r}}_{3D}(t_n)) - \vec{r}_{2D}(t_n)\|_2$$

J.4. Macro F1 Score

J.4。宏平均 F1 分数

For real-world datasets with binary spin labels (topspin/backspin), we compute the Macro F1 score.

对于具有二元旋转标签（上旋/下旋）的真实数据集，我们计算宏 F1 分数。

To obtain a binary class from our network’ s continuous 3D spin prediction $\hat{\vec{\omega}}(t_0)$, we first transform the spin vector from the world coordinate system into the ball coordinate system defined in (Kienzle et al., 2025) .

为了从我们网络的连续 3D 旋转预测中获得二元类$\hat{\vec{\omega}}(t_0)$，我们首先将旋转向量从世界坐标系转换到 (Kienzle et al., 2025) 中定义的球坐标系。

In this local frame, the $\tilde{y}$ -axis is orthogonal to the ball’ s velocity and parallel to the table plane, such that the spin component $\hat{\omega}_{\tilde{y}}$ directly corresponds to the topspin/backspin magnitude.

在此局部坐标系中，$\tilde{y}$轴与球的速度正交并平行于桌面平面，因此旋转分量$\hat{\omega}_{\tilde{y}}$直接对应于上旋/下旋幅度。

We classify the segment as topspin if $\hat{\omega}_{\tilde{y}} > 0$ and as backspin if $\hat{\omega}_{\tilde{y}} \leq 0$.

如果$\hat{\omega}_{\tilde{y}} > 0$，我们将该段分类为上旋；如果$\hat{\omega}_{\tilde{y}} \leq 0$，我们将该段分类为下旋。

We then calculate the F1 score for each class $c \in \{\text{Topspin}, \text{Backspin}\}$:

随后计算各类别 $c \in \{\text{Topspin}, \text{Backspin}\}$ 的 F1 分数：

(SM18) $$\text{F1}_c = \frac{2 \cdot \text{TP}_c}{2 \cdot \text{TP}_c + \text{FP}_c + \text{FN}_c}$$

where $\text{TP}_c, \text{FP}_c, \text{FN}_c$ are the true positives, false positives, and false negatives for class c .

其中$\text{TP}_c, \text{FP}_c, \text{FN}_c$是$c$类的真阳性、假阳性和假阴性。

The final Macro F1 score is the unweighted mean:

最终的 Macro F1 分数是未加权平均值：

(SM19) $$\text{Macro F1} = \frac{\text{F1}_{\text{Topspin}} + \text{F1}_{\text{Backspin}}}{2}$$

Appendix K Additional Experiments and Visualizations

附录 K 附加实验和可视化

K.1. Lifting Network Training

K.1。 三维提升网络训练

The network, consisting of 1.6 million parameters, is trained solely on the train split of our synthetic dataset (2.6 million rallies).

网络含 1.6 百万个参数，只在合成数据集的训练划分上训练，其中包含 2.6 百万个回合。

It is trained for 17 epochs on a single NVIDIA H100 GPU and we use the ADAM optimizer (Kingma and Ba, 2015) with a learning rate of 10^{-4} .

它在单个 NVIDIA H100 GPU 上训练17个周期，我们使用 ADAM 优化器（Kingma 和 Ba，2015），学习率为10^{-4}。

We track an exponential moving average of the model weights (Tarvainen and Valpola, 2017) and select the best model based on its validation performance on the TTST dataset (Kienzle et al., 2026) .

我们跟踪模型权重的指数移动平均值（Tarvainen 和 Valpola，2017），并根据其在 TTST 数据集上的验证性能选择最佳模型（Kienzle 等人，2026）。

K.2. Lifting Network Inference Speed

K.2。 三维提升网络推理速度

Unlike optimization-based approaches (Etaat et al., 2025 ; Gossard et al., 2025 ; Liu and Wang, 2022) , which require a computationally expensive fitting process for every individual segment of a point, learning-based methods (Kienzle et al., 2025 ; Kienzle et al., 2026) perform lifting via a single, efficient forward pass.

与基于优化的方法（Etaat et al., 2025；Gossard et al., 2025；Liu and Wang, 2022）不同，基于学习的方法（Kienzle et al., 2025；Kienzle et al., 2026）通过单个高效的前向传递执行三维提升。

While the initial training cost is non-negligible (approximately 2 days on a single NVIDIA H100 GPU), the resulting model offers exceptional efficiency during inference.

虽然初始训练成本不可忽略（在单个 NVIDIA H100 GPU 上大约需要 2 天），但生成的模型在推理过程中提供了卓越的效率。

Our proposed network further enhances this efficiency by processing entire points consisting of multiple segments in a single forward pass, rather than lifting each segment individually.

本文网络在一次前向传播中处理包含多拍的完整回合，进一步提高效率，无须逐段进行三维提升。

We evaluate the inference speed of our Lifting Network across three generations of GPU hardware: NVIDIA H100, V100, and Titan X (Pascal).

我们在 NVIDIA H100、V100 和 Titan X（Pascal）三代 GPU 上评估三维提升网络的推理速度。

We distinguish between two operating regimes: an online mode (batch size 1), representing latency-critical real-time applications, and an offline mode (batch size 128), representing high-throughput dataset generation.

我们区分两种操作模式：在线模式（批量大小 1），代表延迟关键型实时应用程序；离线模式（批量大小 128），代表高吞吐量数据集生成。

The results are summarized in Table SM2 .

结果总结于表SM2中。

Table SM2 .

表 SM2。

Inference performance of the Lifting Network.

三维提升网络的推理性能。

The network’ s high throughput (measured in Rallies per Second) allows for extremely efficient large-scale dataset processing. ✓/× indicate the presence/absence of batching (batch size 128).

网络吞吐量以每秒处理的回合数衡量，支持高效的大规模数据处理。✓/× 表示是否采用批处理，批量大小为 128。

Batched 批量	Inference Speed (Rallies/s) 推理速度（回合/秒）	GPU VRAM GPU显存	GPU 图形处理器
×	120	650 MB	H100
✓	3011	1070 MB	
×	75	410 MB	V100
✓	1310	810 MB	
×	25	280 MB	Titan X (Pascal) 泰坦 X（帕斯卡）
✓	543	670 MB	

The results demonstrate that our model achieves real-time performance even on legacy hardware.

结果表明，我们的模型即使在传统硬件上也能实现实时性能。

On a Titan X (Pascal), the network processes 25 points per second in online mode.

在 Titan X（Pascal）上，网络以在线模式每秒可处理 25 个回合。

Given that a typical table tennis point lasts several seconds, this inference speed is orders of magnitude faster than real-time, enabling low-latency applications such as live broadcasting analysis or robotic anticipation on consumer-grade hardware.

由于典型回合持续数秒，这一推理速度明显快于视频的实际时长，作者据此提出消费级硬件上的低延迟转播分析或机器人预判应用。

In the batched offline setting, the throughput scales dramatically, reaching over 3000 rallies per second on an H100.

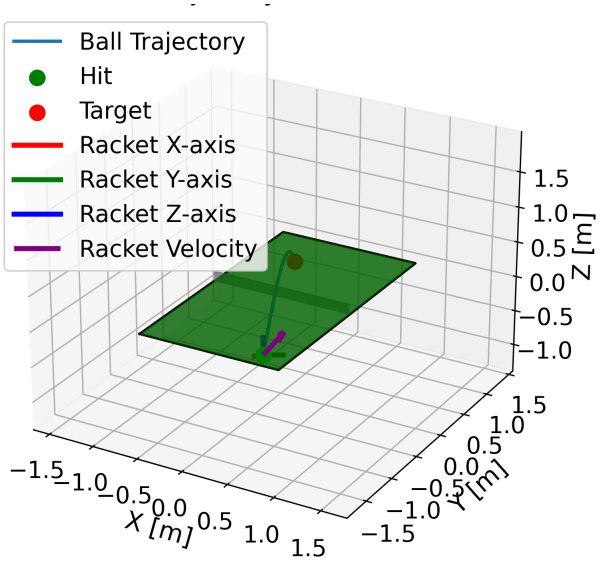
批量离线模式下，H100 的吞吐量可超过每秒 3000 个回合。

This extreme efficiency was a critical enabler for the creation of the TT4D dataset, allowing us to lift hundreds of hours of gameplay in minutes, shifting the computational bottleneck entirely to the preliminary steps.

这种效率使数百小时比赛轨迹的三维提升可在数分钟内完成，将主要计算瓶颈转移至前期处理。

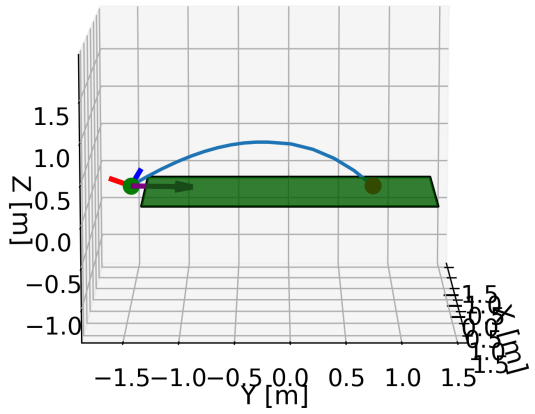
K.3. Racket Strike Reconstruction Visualization

K.3。球拍击球重建可视化



(a) Isometric view

(a) 等角视图



(b) Side view

(b) 侧视图

Figure SM9 .

图SM9。

Example of the estimated racket orientation and velocity for a given ball time of flight, ball incoming velocity and spin and bounce target.

给定球飞行时间、球传入速度以及旋转和弹跳目标的估计球拍方向和速度的示例。

This was solved using our OCP .

使用我们的 OCP 解决了这个问题。

Two 3D plots, an isometric view and a side view, illustrating a table tennis shot.

两个 3D 图（一个等距视图和一个侧视图）展示了乒乓球击球。

A legend in the isometric view identifies a blue line as the ball trajectory, a green dot as the hit, a red dot as the target, and red, green, blue, and purple lines representing the racket X-axis, Y-axis, Z-axis, and velocity, respectively.

等距视图中的图例将蓝线标识为球轨迹，绿点表示击球，红点表示目标，红、绿、蓝和紫色线分别代表球拍的 X 轴、Y 轴、Z 轴和速度。

Both plots show the blue ball trajectory arching from the hit point on one side of a green table plane to the target point on the other, with the racket orientation axes and velocity vectors originating from the hit point.

两张图都显示了从绿色桌面平面一侧的击球点到另一侧目标点的蓝色球轨迹，球拍方向轴和速度矢量源自击球点。

To further validate our racket reconstructions, we provide a qualitative visualization of the reconstruction in Figure SM9 .

为了进一步验证我们的球拍重建，我们在图 SM9 中提供了重建的定性可视化。

The figure shows that the racket reconstructions look realistic.

该图显示球拍重建看起来很真实。

K.4. Evaluation of Generative Data

K.4。生成数据的评估

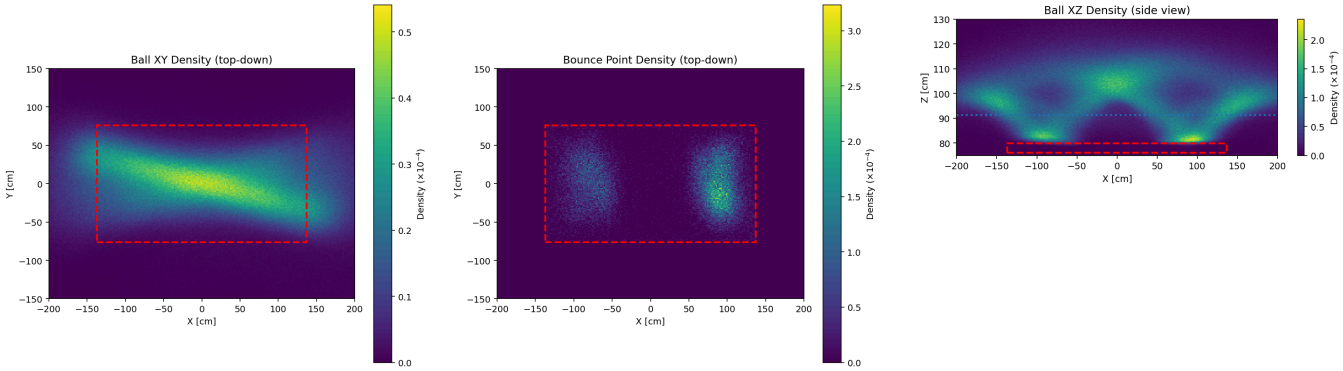


Figure SM10 .

图SM10。

Ball position densities for the 10,000 generated samples.

生成的 10,000 个样本的球位置密度。

The table region is marked by the dashed red line, and the net’ s height is marked by the dotted blue line.

球台区域由红色虚线标记，球网的高度由蓝色虚线标记。

Three heatmaps illustrating ball position and bounce densities for 10,000 generated samples.

三幅热图说明了 10,000 个生成样本的球位置和弹跳密度。

A dashed red rectangle marks the table boundaries in all plots.

红色虚线矩形标记了所有图中的球台边界。

The top-left plot displays a top-down view of ball XY density, revealing a prominent diagonal band indicative of frequent cross-court trajectories.

左上图显示了球 XY 密度的自上而下视图，显示了一条显着的对角线带，表明频繁的斜线轨迹。

The top-right plot shows top-down bounce point density, featuring two distinct high-density clusters on opposite halves of the table.

右上图显示了自上而下的弹跳点密度，在球台的相对两半上有两个不同的高密度簇。

The bottom plot presents a side view of ball XZ density, showing the ball's trajectory arching low over a dotted blue line representing the net, with density concentrating near the table surface on both sides indicating bounces.

底部图显示了球 XZ 密度的侧视图，显示了球的轨迹在代表球网的蓝色虚线上呈低拱形，密度集中在两侧的桌面附近，表示弹跳。

We provide a qualitative example of a generated sequence in Figure SM12 .

我们在图 SM12 中提供了生成序列的定性示例。

Moreover, we illustrate the distribution of the ball locations in Figure SM10 .

此外，我们在图 SM10 中说明了球位置的分布。

K.5. Humanoid Motion Tracking of Table Tennis Motions

K.5。乒乓球运动的人形机器人动作跟踪

We validate the fidelity of our Player Reconstruction algorithm (A.3) by replaying the motion on a Unitree G1 robot.

我们通过在 Unitree G1 机器人上重放运动来验证球员重建算法 (A.3) 的保真度。

First, we retarget the SMPL motion to the Unitree G1 using GMR (Araujo et al., 2026) .

首先使用 GMR（Araujo 等，2026），将 SMPL 人体动作重定向至 Unitree G1。

Next, to facilitate smooth hardware deployment, we use the motion in-betweening capabilities of GEM (Li et al., 2025) to ease the start and end of each motion.

为使硬件部署时动作平滑，使用 GEM 的动作过渡生成能力（Li 等，2025）处理每段动作的起止部分。

Specifically, we generate half a second of motion to smoothly transition from an A-pose to the initial motion pose and from the final motion pose back to an A-pose.

具体来说，我们生成半秒的运动，以平滑地从 A 姿态过渡到初始运动姿态，并从最终运动姿态平滑地过渡回 A 姿态。

Finally, we train a motion tracking policy on this smoothed motion using BeyondMimic (Liao et al., 2025) for 30,000 iterations on an NVIDIA GeForce RTX 5090 GPU.

最后，我们使用 BeyondMimic（Liao 等人，2025）在 NVIDIA GeForce RTX 5090 GPU 上训练这种平滑运动的运动跟踪策略，进行 30,000 次迭代。

One hardware deployment video along with the original motion is included in the supplementary zip file.

补充 zip 文件中包含一段硬件部署视频以及原始动作。

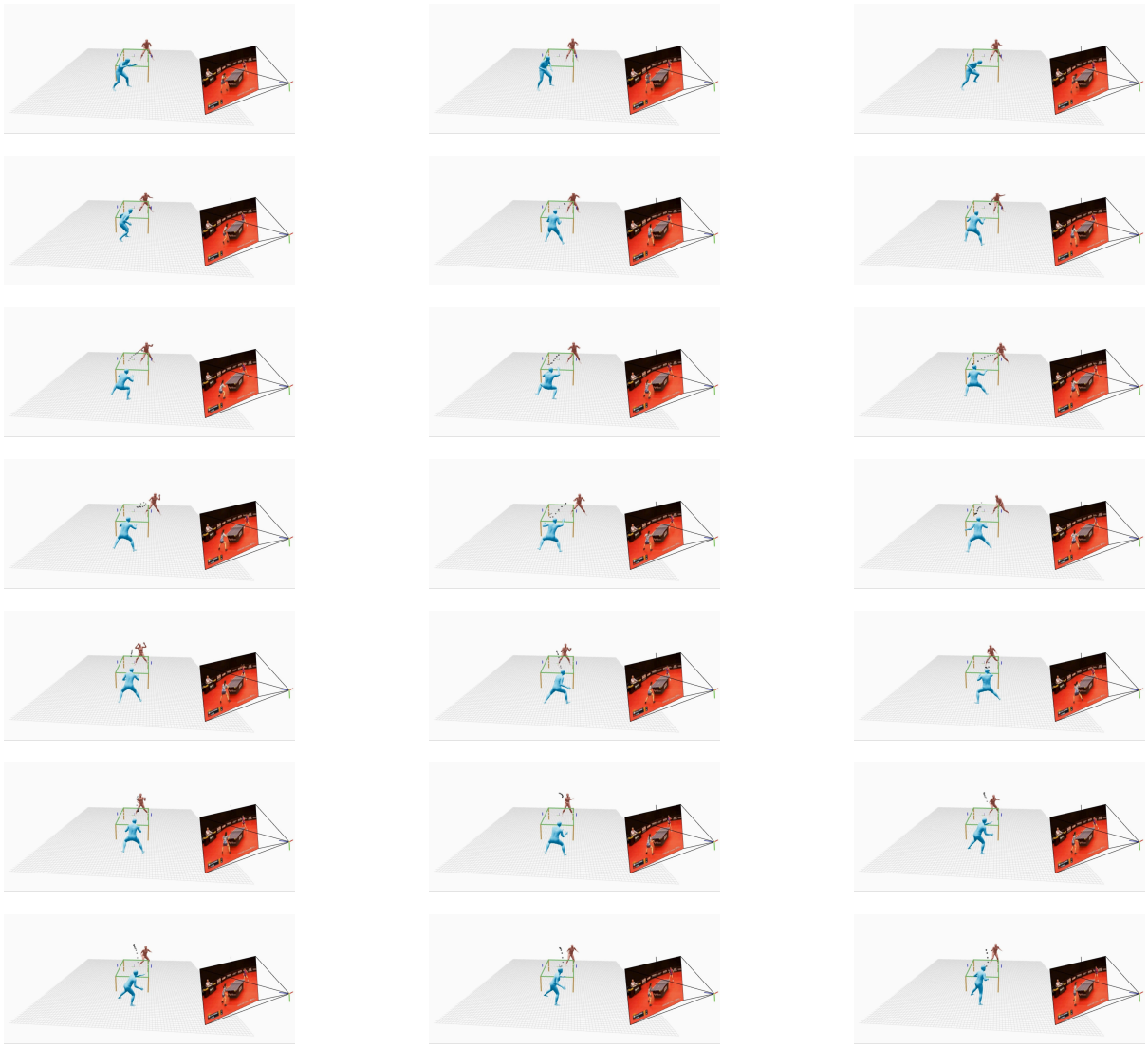


Figure SM11 .

图SM11。

One example of a reconstructed sequence from our TT4D dataset at 30 FPS.

从我们的 TT4D 数据集以 30 FPS 重建序列的一个示例。

We display every 16th frame.

我们每 16 帧显示一次。

The diagram should be read from left to right.

该图应从左到右阅读。

A 7 by 3 grid of frames showing a chronological sequence of a reconstructed table tennis rally.

7 x 3 的帧网格显示了重建的乒乓球回合的时间顺序。

Each frame displays a 3D scene with two human meshes (one light blue in the foreground, one red in the background) playing at a green wireframe table, with the ball's trajectory tracked by a series of dots.

每个帧都显示一个 3D 场景，其中有两个人体网格（前景为浅蓝色，背景为红色）在绿色线框球台上玩耍，球的轨迹由一系列点跟踪。

On the right side of each frame, a 2D inset of the original broadcast video is projected from a camera frustum, illustrating the estimated camera viewpoint.

在每帧的右侧，原始转播视频的 2D 插图从摄像机平截头体投影，说明了估计的摄像机视点。

The sequence progresses from left to right, top to bottom, showing the continuous player movements and ball exchanges.

该序列从左到右、从上到下进行，显示连续的球员动作和球交换。



Figure SM12 .

图SM12。

One example generated sequence at 30 FPS.

一个示例以 30 FPS 生成序列。

We display every 16th frame.

我们每 16 帧显示一次。

The diagram should be read from left to right.

该图应从左到右阅读。

The motion of the skeleton and ball is smooth and realistic.

骨架和球的运动平滑逼真。

A 6 by 4 grid of frames showing a chronological sequence of a generated table tennis rally.

6 x 4 的帧网格显示了生成的乒乓球回合的时间顺序。

Each frame displays a 3D scene featuring two human skeletons rendered in light blue, positioned on opposite sides of a green wireframe table.

每个帧都显示一个 3D 场景，其中有两个以浅蓝色渲染的人体骨骼，位于绿色线框球台的两侧。

A red dot represents the table tennis ball.

红点代表乒乓球。

Reading from left to right and top to bottom, the sequence illustrates the continuous, simulated movement of the skeletal players and the trajectory of the ball back and forth across the net.

从左到右、从上到下阅读，该序列展示了骨骼球员的连续模拟运动以及球在网内来回的轨迹。

Footnotes

脚注

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